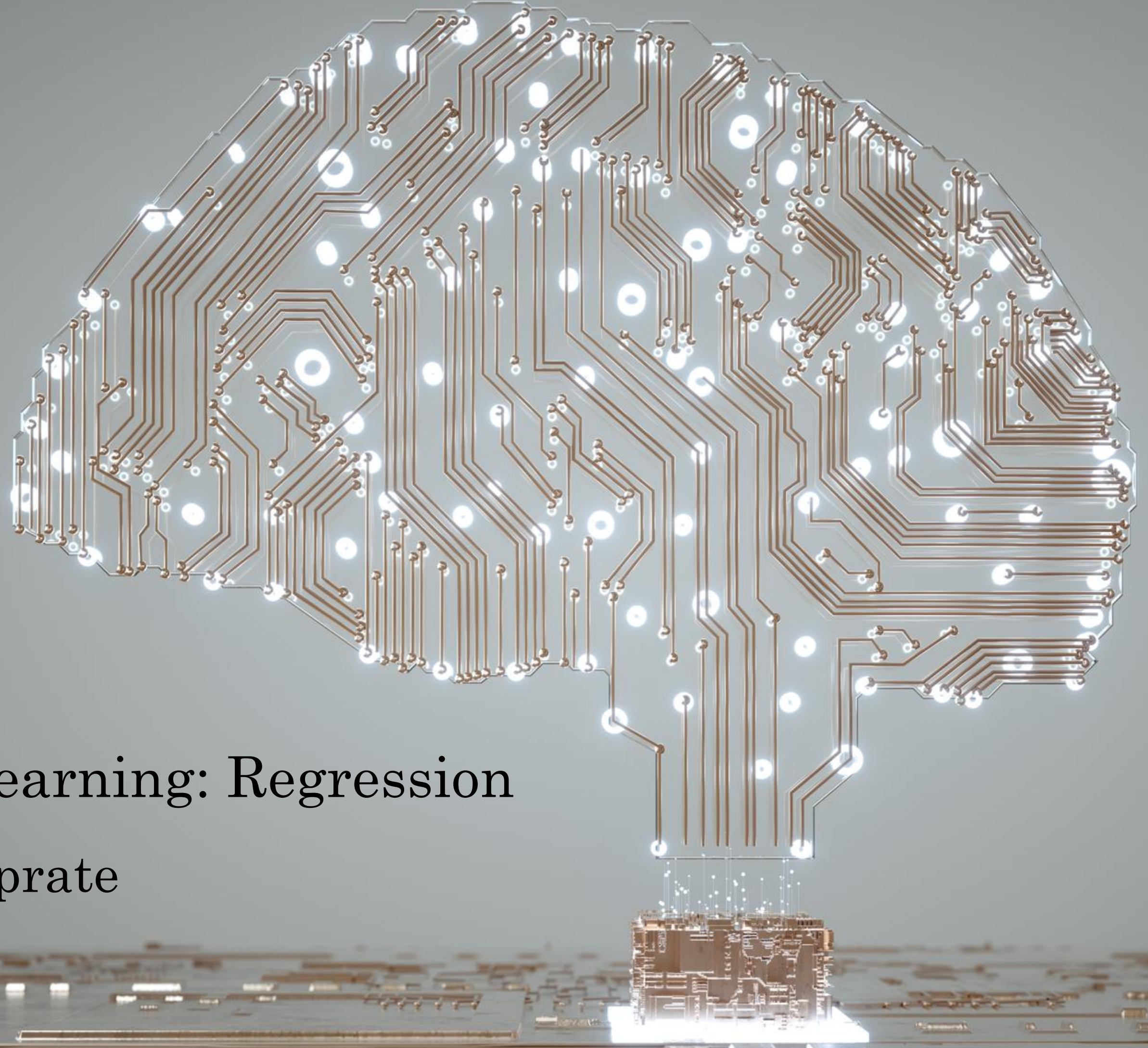
ML & NLP: PGR210



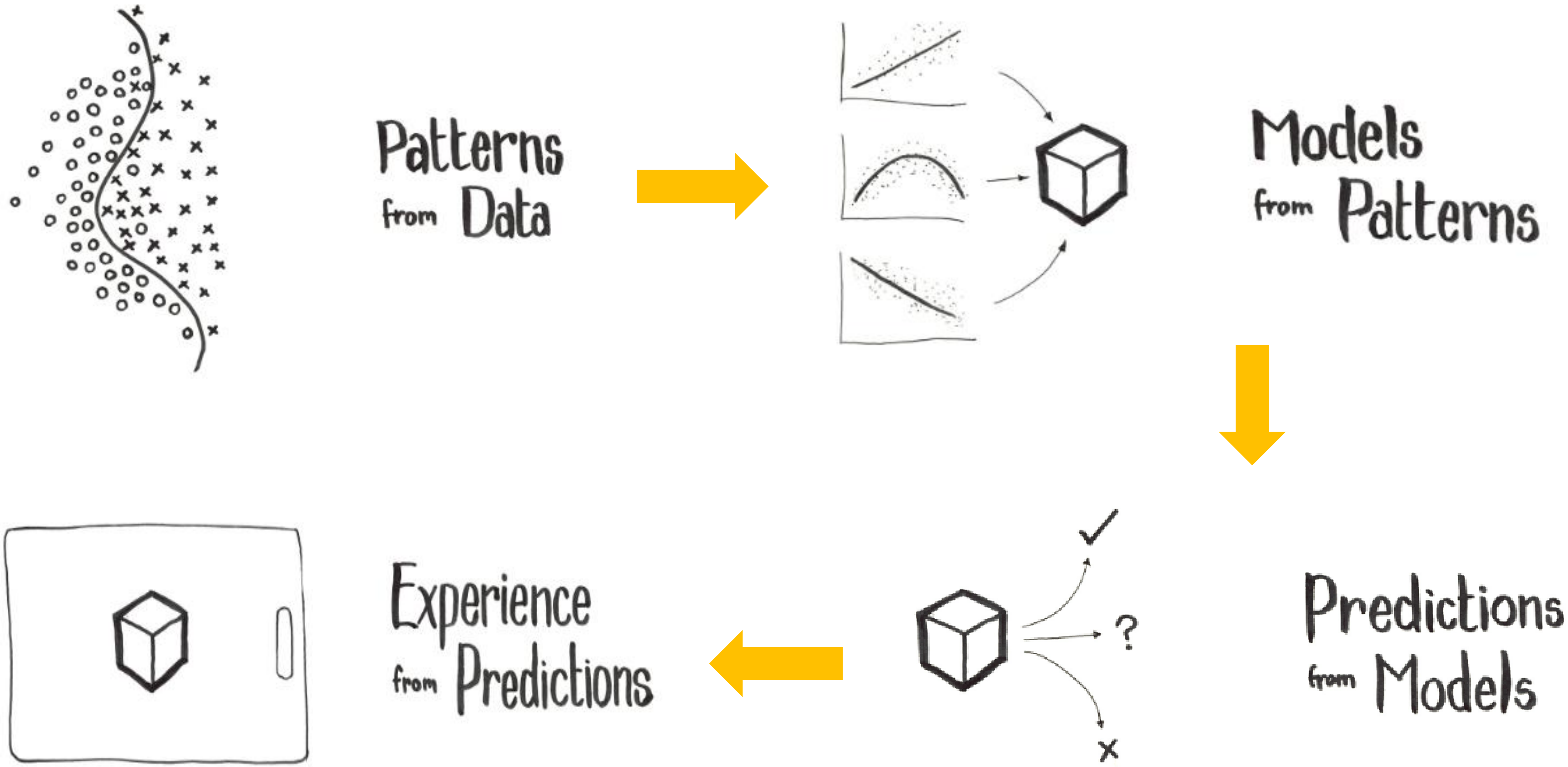
Supervised Learning: Regression

Dr. Arvind Keprate

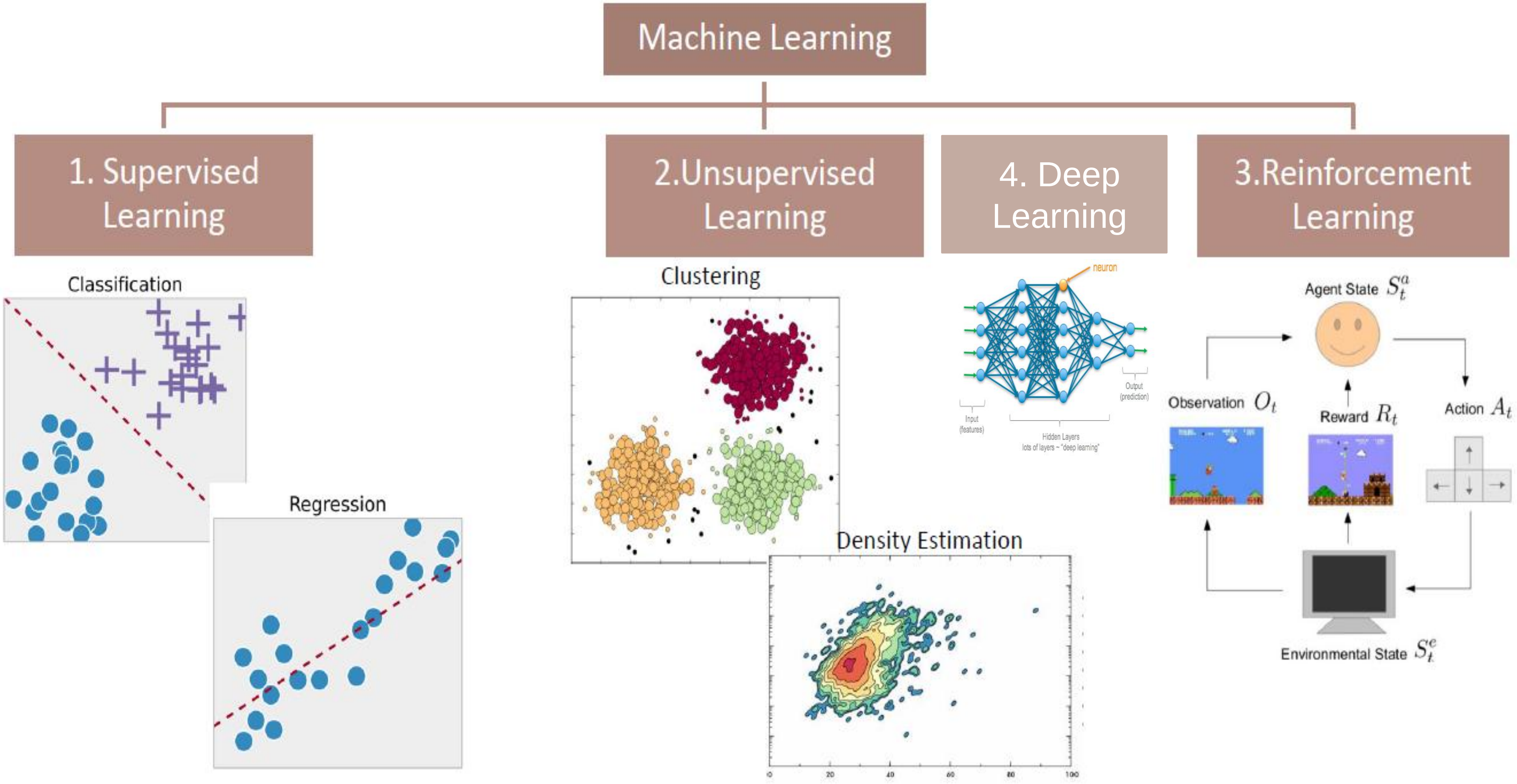
Topics for Today

- Introduction
- What is Linear Regression?
- Model Evaluation
- Evaluation Metrics
- Multi-Linear Regression
- Polynomial Regression
- Regularization

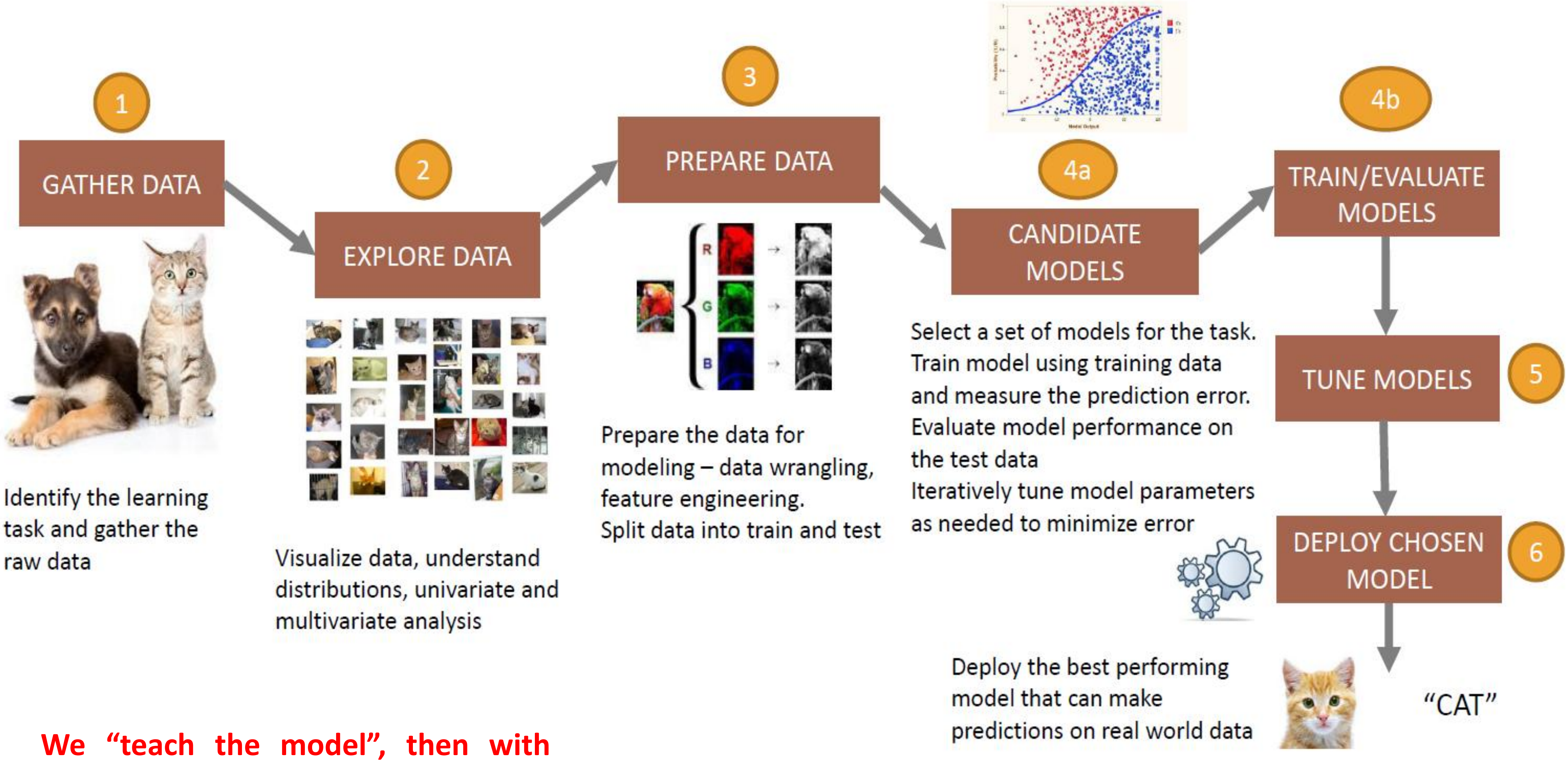
Basic Intuition of ML



Types of ML?

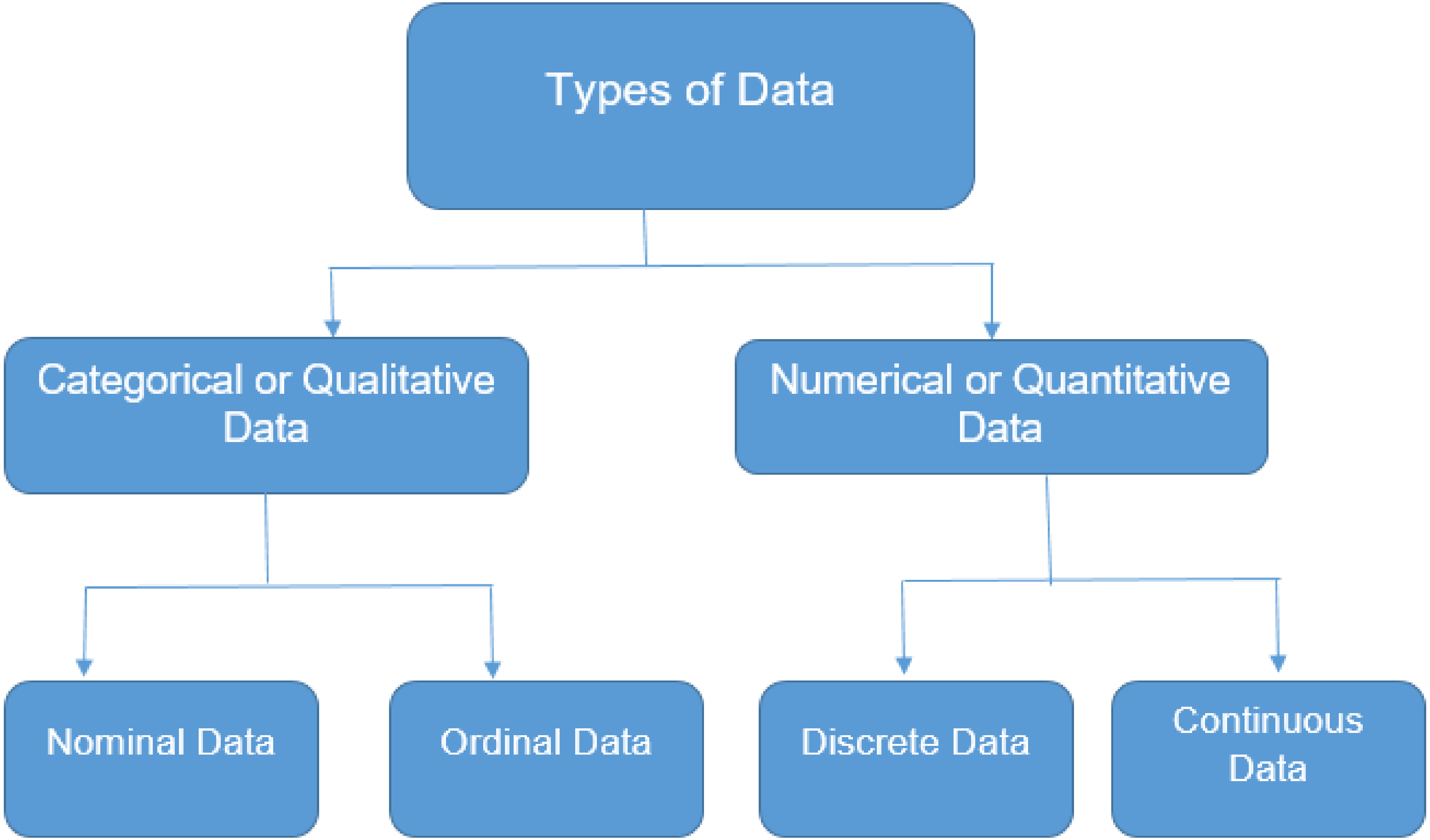

Machine Learning
1. Supervised Learning
2.Unsupervised Learning
4. Deep Learning
3.Reinforcement Learning
Classification
Regression
Clustering
Density Estimation
neuron
Input (features)
Hidden Layers
lots of layers ~ "deep learning"
Output (prediction)
Agent State S_t^a
Observation O_t
Reward R_t
Action A_t
Environmental State S_t^e

Supervised ML Workflow



We "teach the model", then with that knowledge, it can predict unknown or future instances.

Types of Data



Sample Structured Data

Column Names = Attributes

ID	Clump	UnifSize	UnifShape	MargAdh	SingEpiSize	BareNuc	BlandChrom	NormNucl	Mit	Class
1000025	5	1	1	1	2	1	3	1	1	benign
1002945	5	4	4	5	7	10	3	2	1	benign
1015425	3	1	1	1	2	2	3	1	1	malignant
1016277	6	8	8	1	3	4	3	7	1	benign
1017023	4	1	1	3	2	1	3	1	1	benign
1017122	8	10	10	8	7	10		7	1	malignant
1018099	1	1	1	1	2	10	3	1	1	benign
1018561	2	1	2	H	2	1	3	1	1	benign
1033078	2	1	1	1	2	1	1	1	5	benign
1033078	4	2	1	1	2	1	2	1	1	benign

Row = Observation

Columns = Feature

Sample Unstructured Data

Python Libraries for Machine Learning

pandas

scikit learn

NumPy

SciPy

matplotlib

python

scikit-learn

- Free software machine learning library
- Classification, Regression and Clustering algorithms
- Works with NumPy and SciPy
- Great documentation
- Easy to implement

1. Most of the tasks that need to be done in a machine learning pipeline are
2. implemented already in Scikit Learn including pre-processing of data,
3. feature selection, feature extraction, train test splitting,
4. defining the algorithms, fitting models,

scikit-learn functions

Data Preprocessing

```
from sklearn import preprocessing
X = preprocessing.StandardScaler().fit(X).transform(X)
```

Train/Test split

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.33)
```

Algorithm setup

```
from sklearn import svm
clf = svm.SVC(gamma=0.001, C=100.)
```

Model fitting

```
clf.fit(X_train, y_train)
```

scikit-learn functions

Prediction

```
clf.predict(X_test)
```

Evaluation

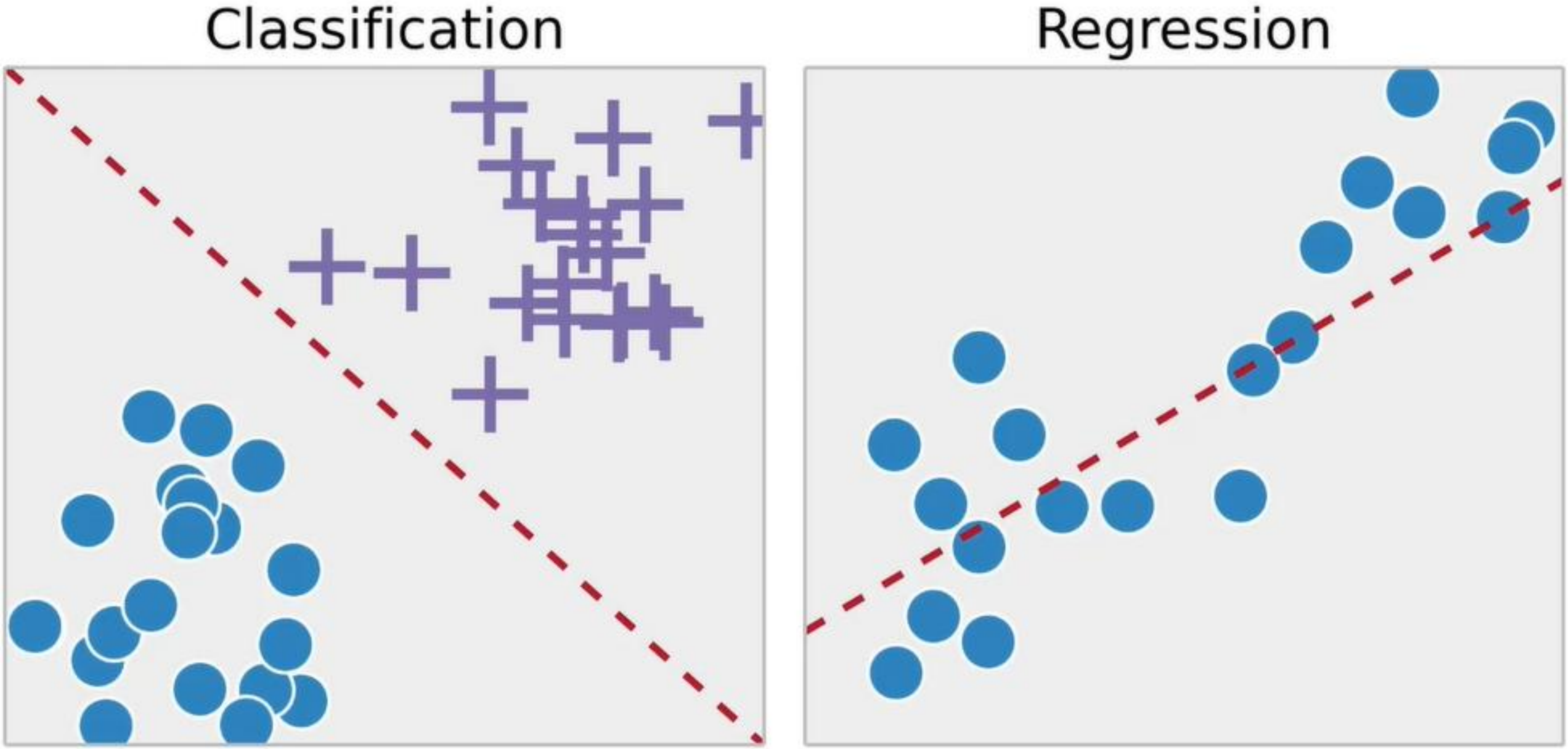
```
from sklearn.metrics import confusion_matrix
print(confusion_matrix(y_test, yhat, labels=[1,0]))
```

Model export

```
import pickle
s = pickle.dumps(clf)
```

ML Performed in 12 Lines of Code ☺.

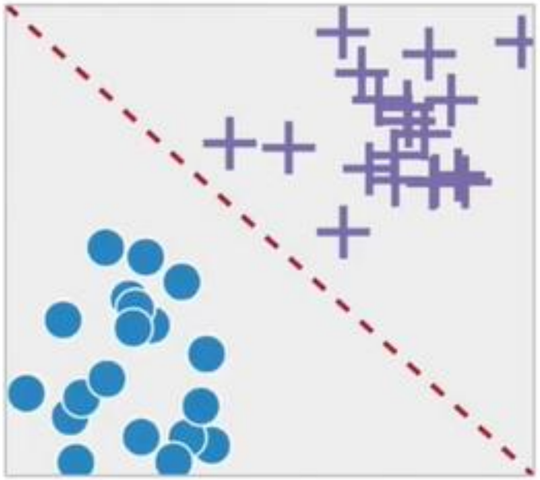
Types of Supervised Learning



What is Classification?

Classification is the process of predicting discrete class labels or categories.

ID	Clump	UnifSize	UnifShape	MargAdh	SingEpiSize	BareNuc	BlandChrom	NormNucl	Mit	Class
1000025	5	1	1	1	2	1	3	1	1	benign
1002945	5	4	4	5	7	10	3	2	1	benign
1015425	3	1	1	1	2	2	3	1	1	malignant
1016277	6	8	8	1	3	4	3	7	1	benign
1017023	4	1	1	3	2	1	3	1	1	benign
1017122	8	10	10	8	7	10		7	1	malignant
1018099	1	1	1	1	2	10	3	1	1	benign
1018561	2	1	2	H	2	1	3	1	1	benign
1033078	2	1	1	1	2	1	1	1	5	benign
1033078	4	2	1	1	2	1	2	1	1	benign

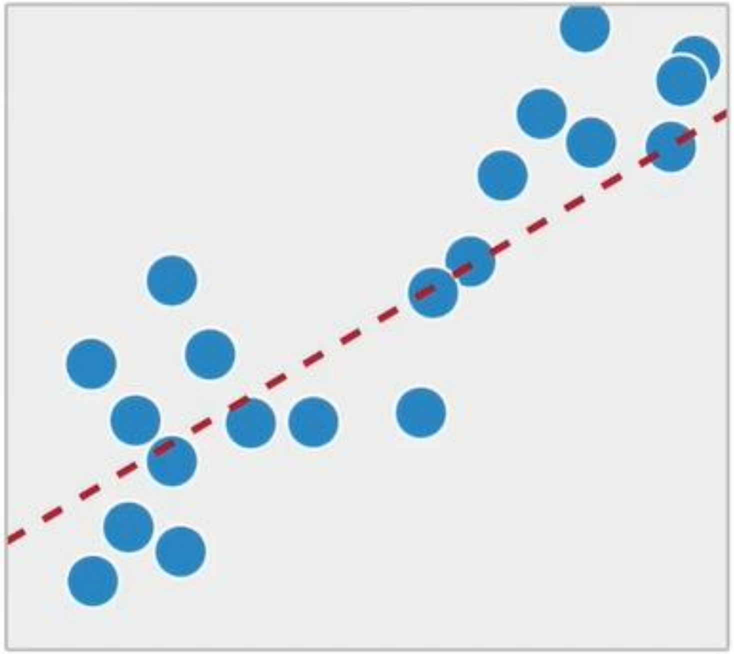


What is Regression?

Regression is the process of predicting continuous values.

	ENGINESIZE	CYLINDERS	FUELCONSUMPTION_COMB	CO2EMISSIONS
0	2.0	4	8.5	196
1	2.4	4	9.6	221
2	1.5	4	5.9	136
3	3.5	6	11.1	255
4	3.5	6	10.6	244
5	3.5	6	10.0	230
6	3.5	6	10.1	232
7	3.7	6	11.1	255
8	3.7	6	11.6	267
9	2.4	4	9.2	?

Continuous Values



What is Unsupervised Learning?

We do not supervise the model, but we let the model work on its own to discover information.

Unsupervised Algorithm trains on the dataset and draws conclusions on unlabeled data.

Customer Id	Age	Edu	Years Employed	Income	Card Debt	Other Debt	Address	DebtIncomeRatio
1	41	2	6	19	0.124	1.073	NBA001	6.3
2	47	1	26	100	4.582	8.218	NBA021	12.8
3	33	2	10	57	6.111	5.802	NBA013	20.9
4	29	2	4	19	0.681	0.516	NBA009	6.3
5	47	1	31	253	9.308	8.908	NBA008	7.2
6	40	1	23	81	0.998	7.831	NBA016	10.9
7	38	2	4	56	0.442	0.454	NBA013	1.6
8	42	3	0	64	0.279	3.945	NBA009	6.6
9	26	1	5	18	0.575	2.215	NBA006	15.5
10	47	3	23	115	0.653	3.947	NBA011	4
11	44	3	8	88	0.285	5.083	NBA010	6.1
12	34	2	9	40	0.374	0.266	NBA003	1.6

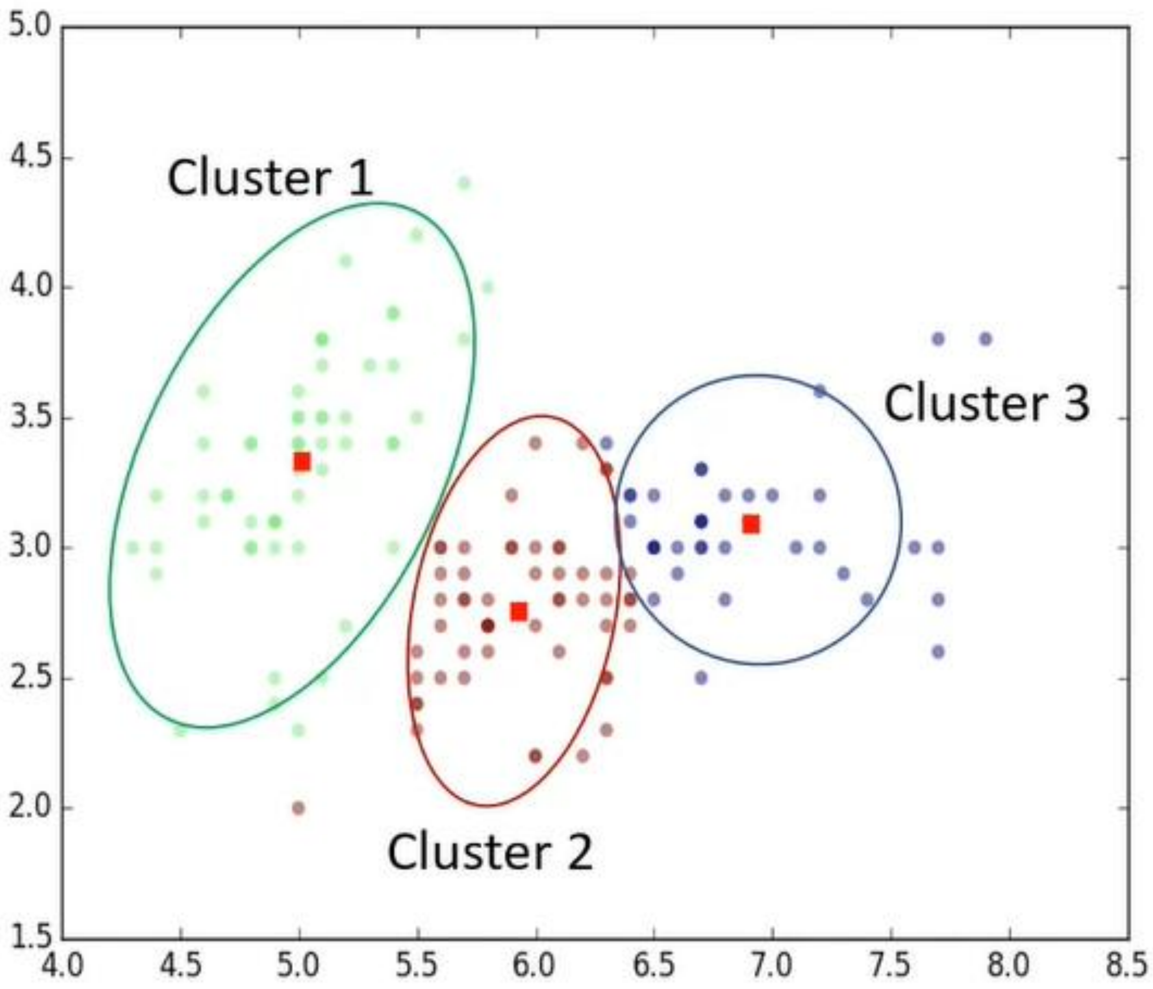
Unsupervised Learning Techniques:

1. Dimension Reduction
2. Density Estimation
3. Market Basket Analysis
4. Clustering

What is Unsupervised Learning?

Clustering is grouping of data points or objects that are somehow similar by:

- Discovering structure
- Summarization
- Anomaly Detection



Supervised vs. Unsupervised Learning

Supervised Learning

- **Classification:**
  Classifies labeled data
- **Regression:**
  Predicts trends using previous labeled data
- Has more evaluation methods than unsupervised learning
- Controlled environment

Unsupervised Learning

- **Clustering:**
  Finds patterns and groupings from unlabeled data
- Has fewer evaluation methods than supervised learning
- Less controlled environment

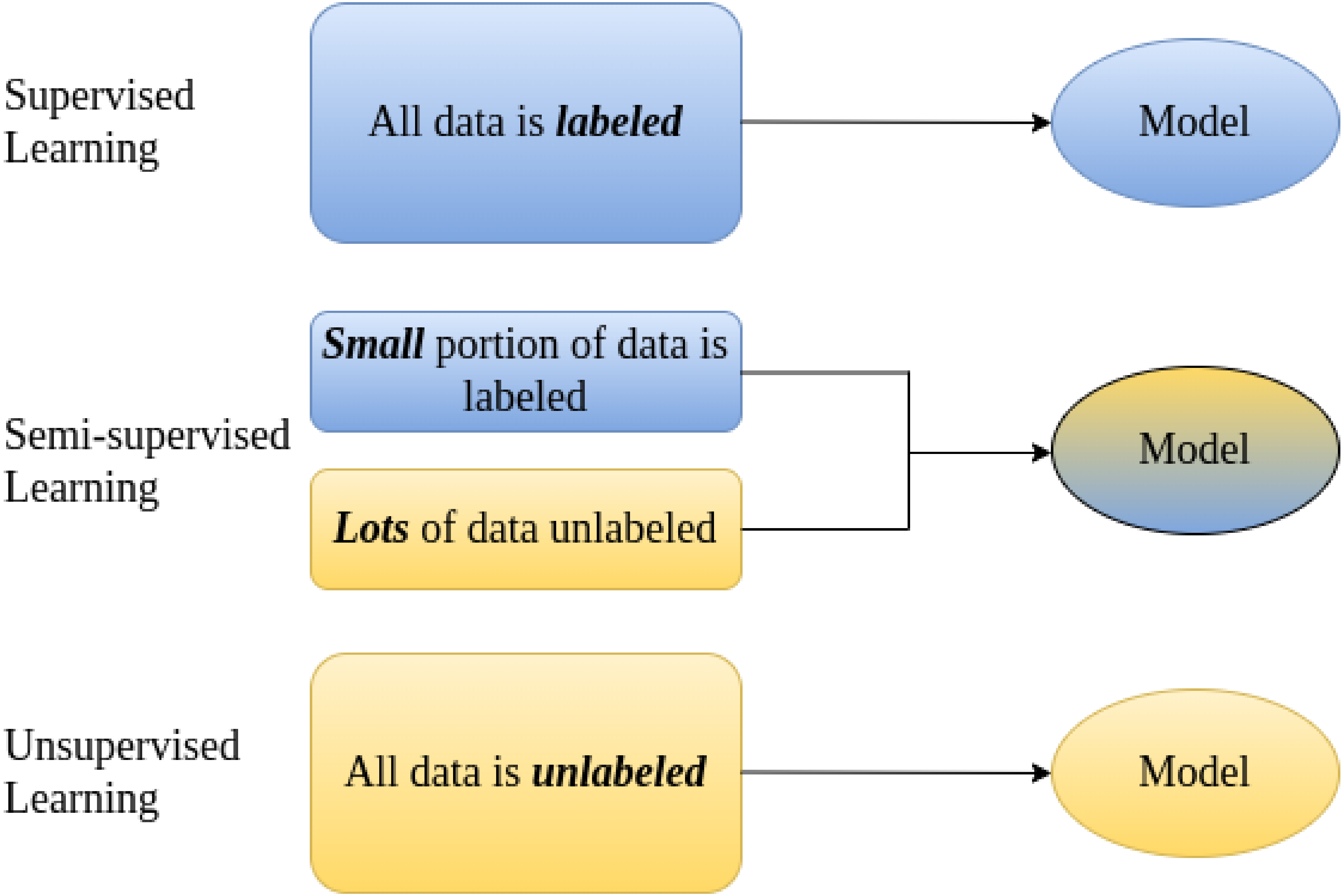
BREAK
15MINS

Reality is bitter?

Most of the Data that is being generated is unlabeled.

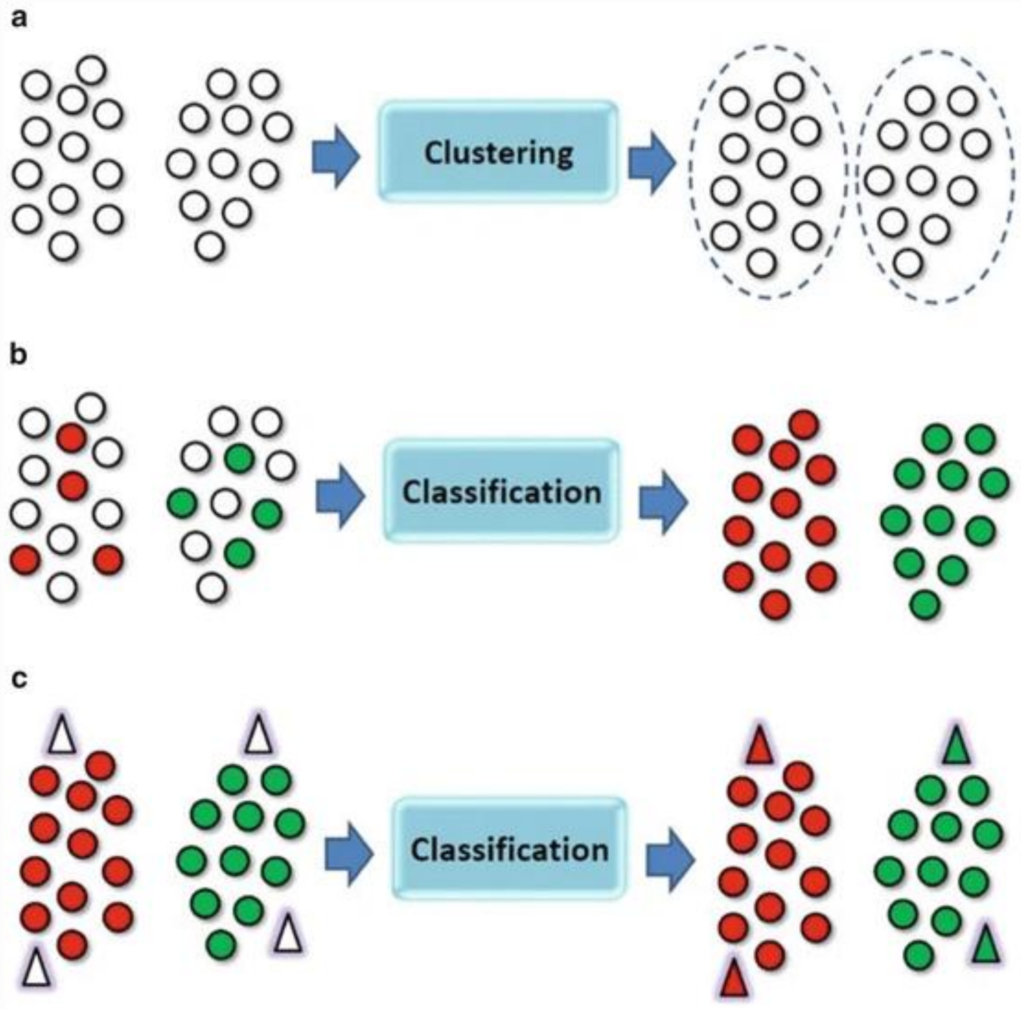
Labeling Data is good, but it is very expensive and time-consuming.

Best of the Both Worlds



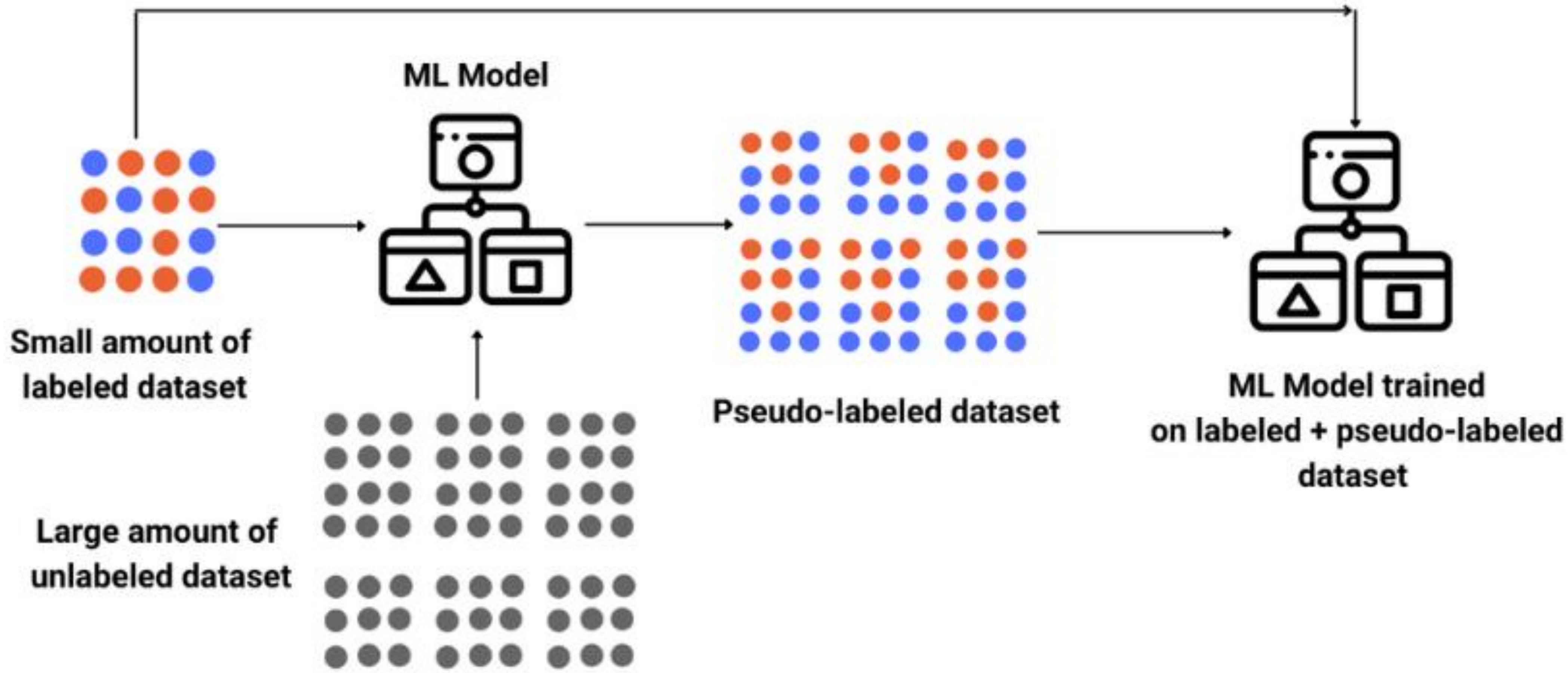
Best of the Both Worlds

Fig. 3.1



Schematic of the three paradigms of the machine learning area. (**a**) Unsupervised learning; (**b**) Semi-supervised learning; (**c**) Supervised learning

What is Semi-Supervised Learning?



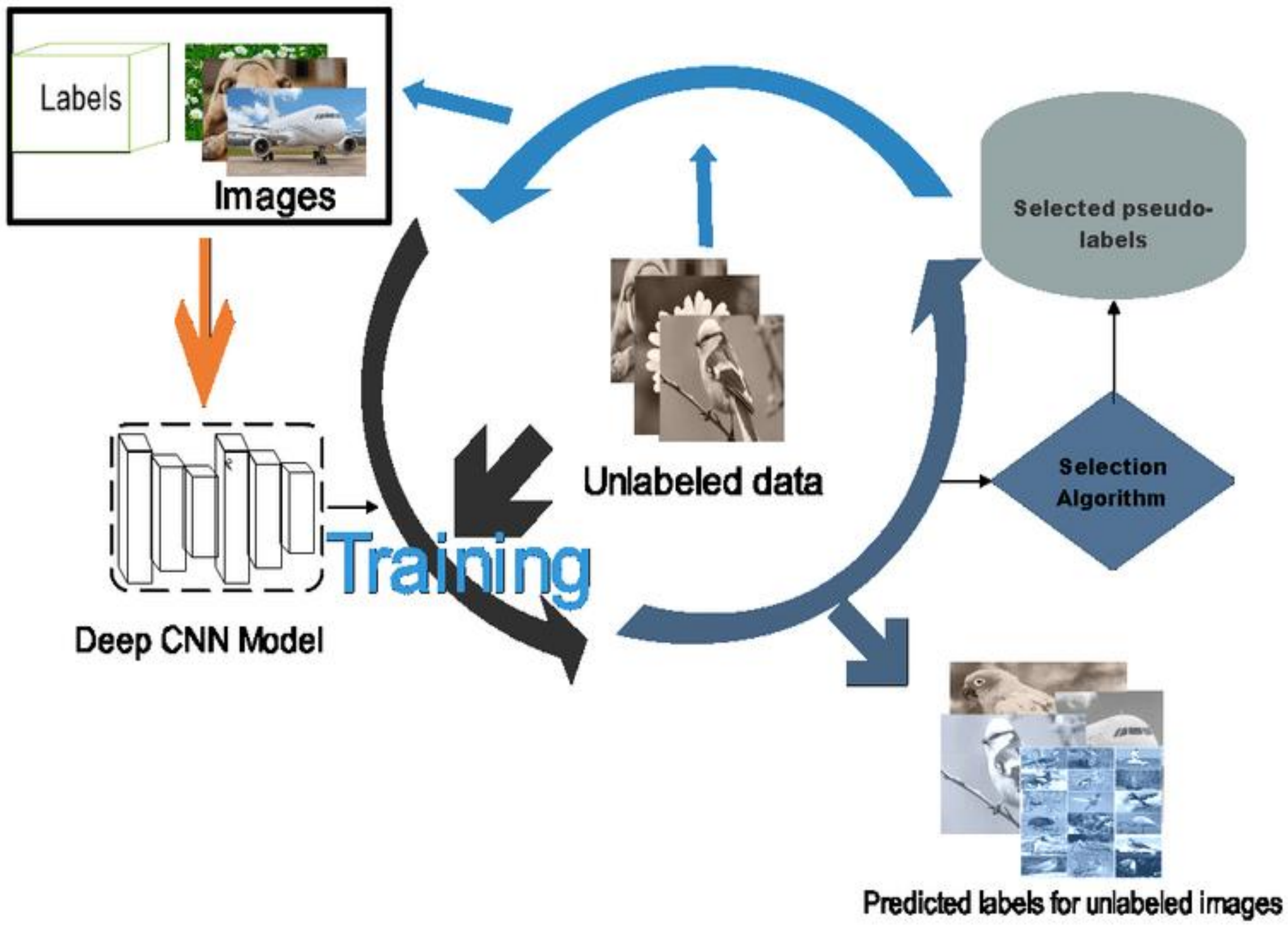
What is Semi-supervised Learning?

- In computer vision, deep learning has shown promising performances beyond human cognitive abilities by learning from large volumes of labeled datasets.

- However, when the labeling of data is expensive or laborious, deep learning is difficult to use effectively because of its relatively poor and unstable performance.

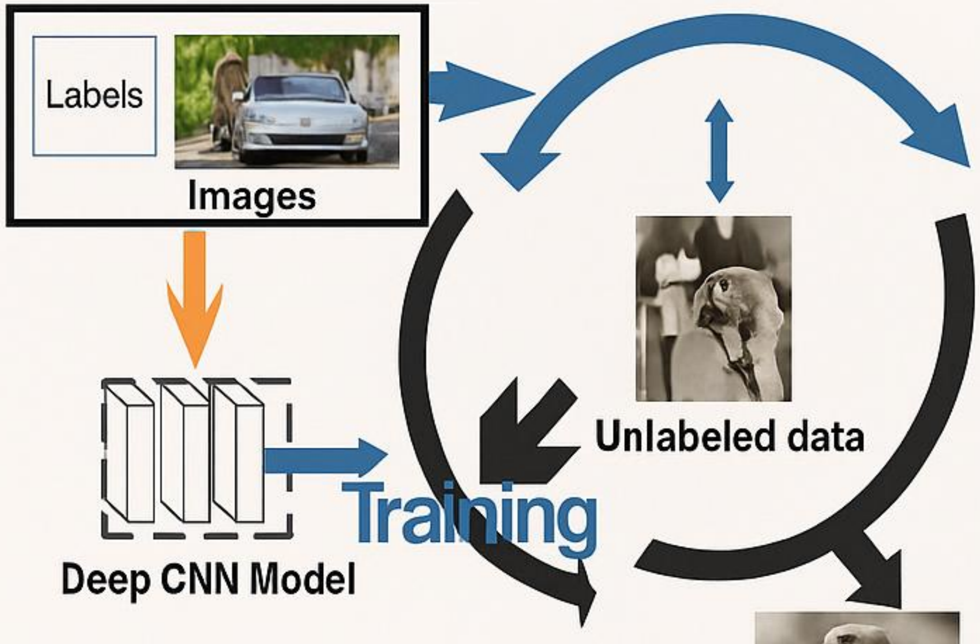
- Semi-supervised learning (SSL) is a practical method that leads to good performance by leveraging a large amount of unlabeled data.

- Since unlabeled data is abundant, easy to get, and inexpensive, semi-supervised learning finds many applications, while the accuracy of results doesn't suffer.



Semi-supervised Learning

Learns from both labeled and unlabeled images

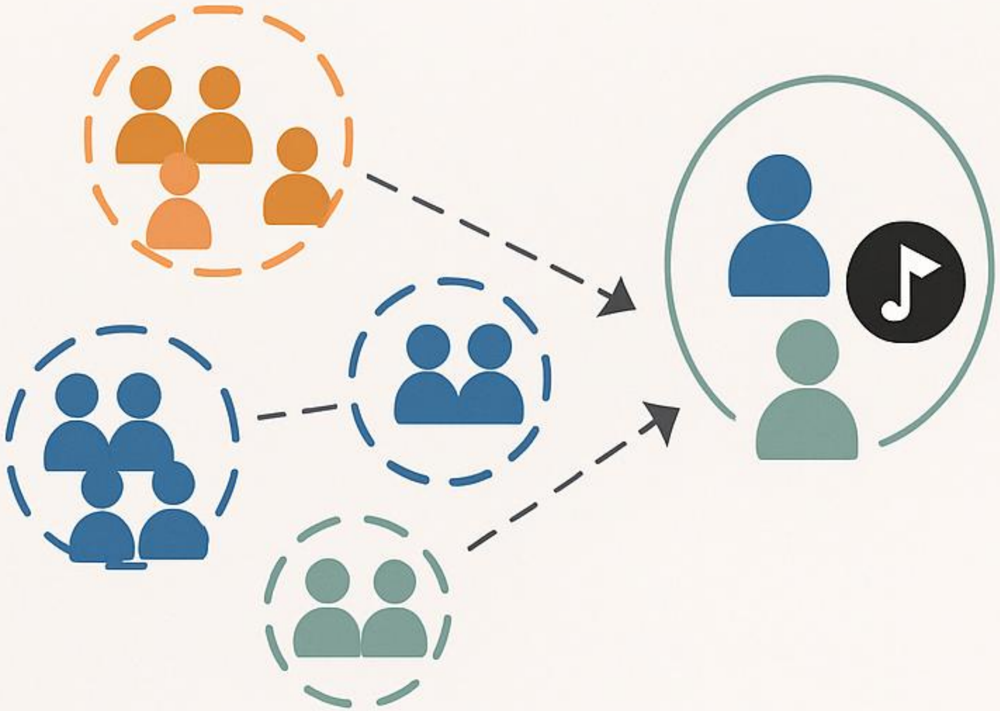


A deep learning model leverages a large amount of unlabeled images to improve accuracy

Unsupervised Learning

(e.g. Spotify/Netflix Recommendations)

Groups similar users to recommend songs or movies



Groups similar users to recommend songs or movies

Unsupervised Learning Algorithms

Clustering

- K-Means
- Polynomial
- Hierarchical
- Fuzzy C-Means

Dimensionality Reduction

- Principal Component Analysis
- Kernel Principal Analysis

Association (Data Mining)

- Apriori Algorithm
- Eclat Algorithm
- FP-Growth Algorithm

Trivia Time

Suppose an online shoe store wants to create a supervised ML model that will provide personalized shoe recommendations to users. That is, the model will recommend certain pairs of shoes to Marty and different pairs of shoes to Janet. The system will use past user behavior data to generate training data. Which of the following statements are true?

- "Shoe beauty" is a useful feature.
- "Shoe size" is a useful feature.
- "The user clicked on the shoe's description" is a useful label.
- "Shoes that a user adores" is a useful label.

Trivia Time

Suppose you want to develop a supervised machine learning model to predict whether a given email is "spam" or "not spam." Which of the following statements are true?

- The labels applied to some examples might be unreliable.
- Emails not marked as "spam" or "not spam" are unlabeled examples.
- We'll use unlabeled examples to train the model.
- Words in the subject header will make good labels.

What is Regression?

- A model is first created based on the data distribution.
- The model is then used to predict future or unknown values.
- Most common approach is **regression analysis:** model the relationship between one or more *independent* or **predictor** variables and a *dependent* or **response** variable.

X: Independent variable Y: Dependent variable

	ENGINESIZE	CYLINDERS	FUELCONSUMPTION_COMB	CO2EMISSIONS
0	2.0	4	8.5	196
1	2.4	4	9.6	221
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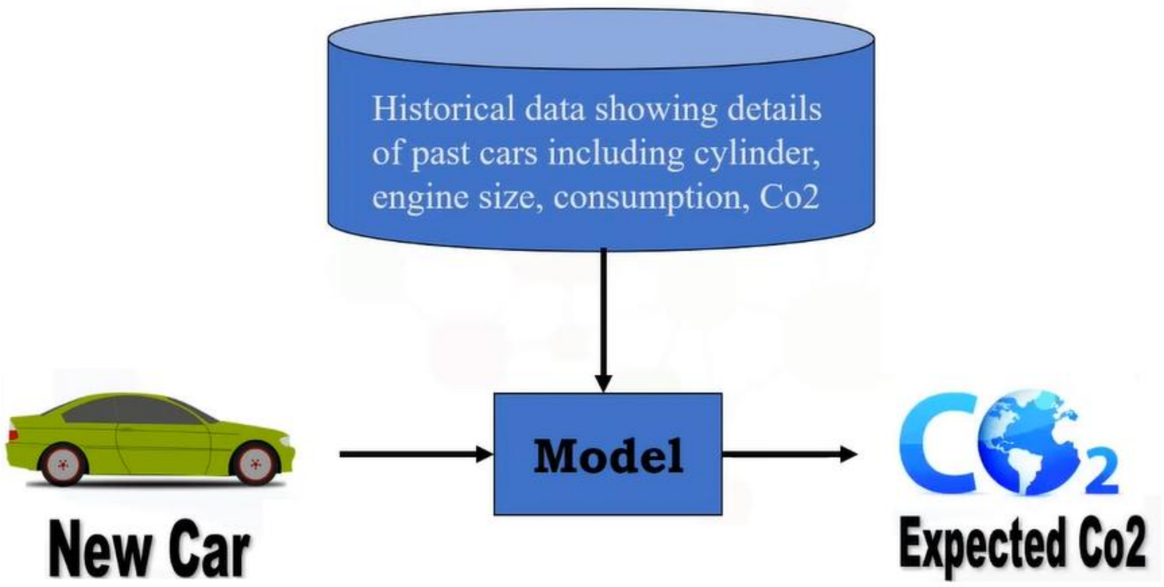
9	2.4	4	9.2	?

Continuous Values



Simple Linear Regression

- Simple Linear Regression:
  - Predict co2emission vs EngineSize of all cars
    - Independent variable (x): EngineSize
    - Dependent variable (y): co2emission

- Multiple Linear Regression:
  - Predict co2emission vs EngineSize and Cylinders of all cars
    - Independent variable (x): EngineSize, Cylinders, etc
    - Dependent variable (y): co2emission

X: Independent variable

Y: Dependent variable

	ENGINESIZE	CYLINDERS	FUELCONSUMPTION_COMB	CO2EMISSIONS
0	2.0	4	8.5	196
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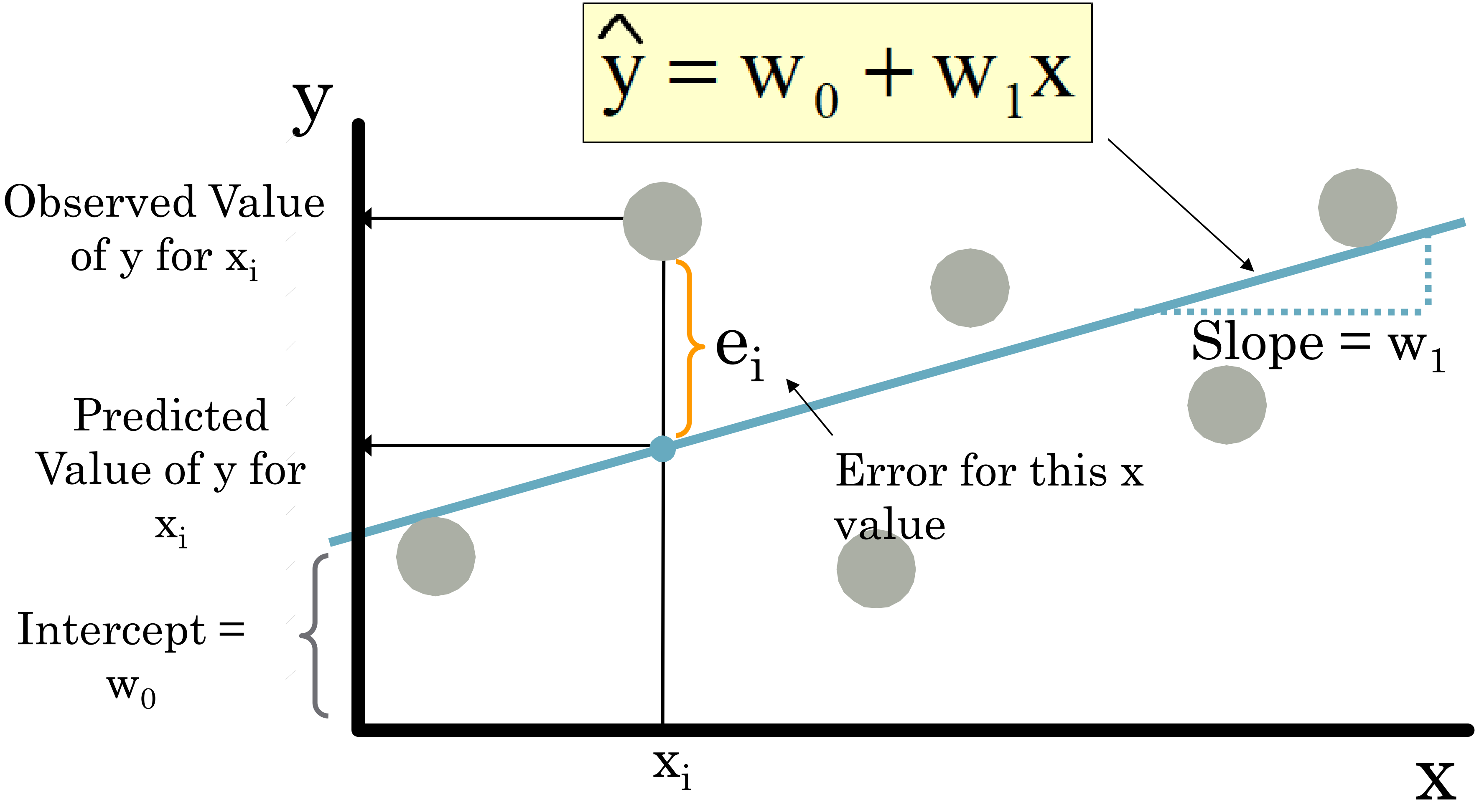
8	3.7	6	11.6	267
9	2.4	4	9.2	?

Continuous Values

How Does Simple Linear Regression Work?



How Does Simple Linear Regression Work?

	ENGINESIZE	CYLINDERS	FUELCONSUMPTION_COMB	CO2EMISSIONS
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1	2.4	4	9.6	221
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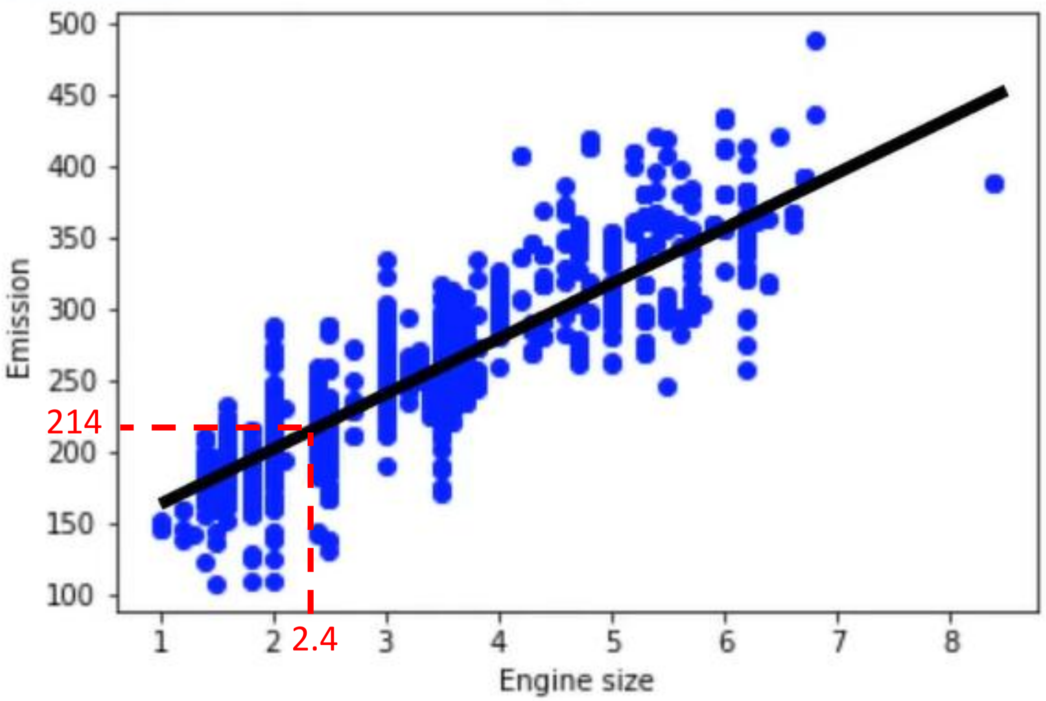


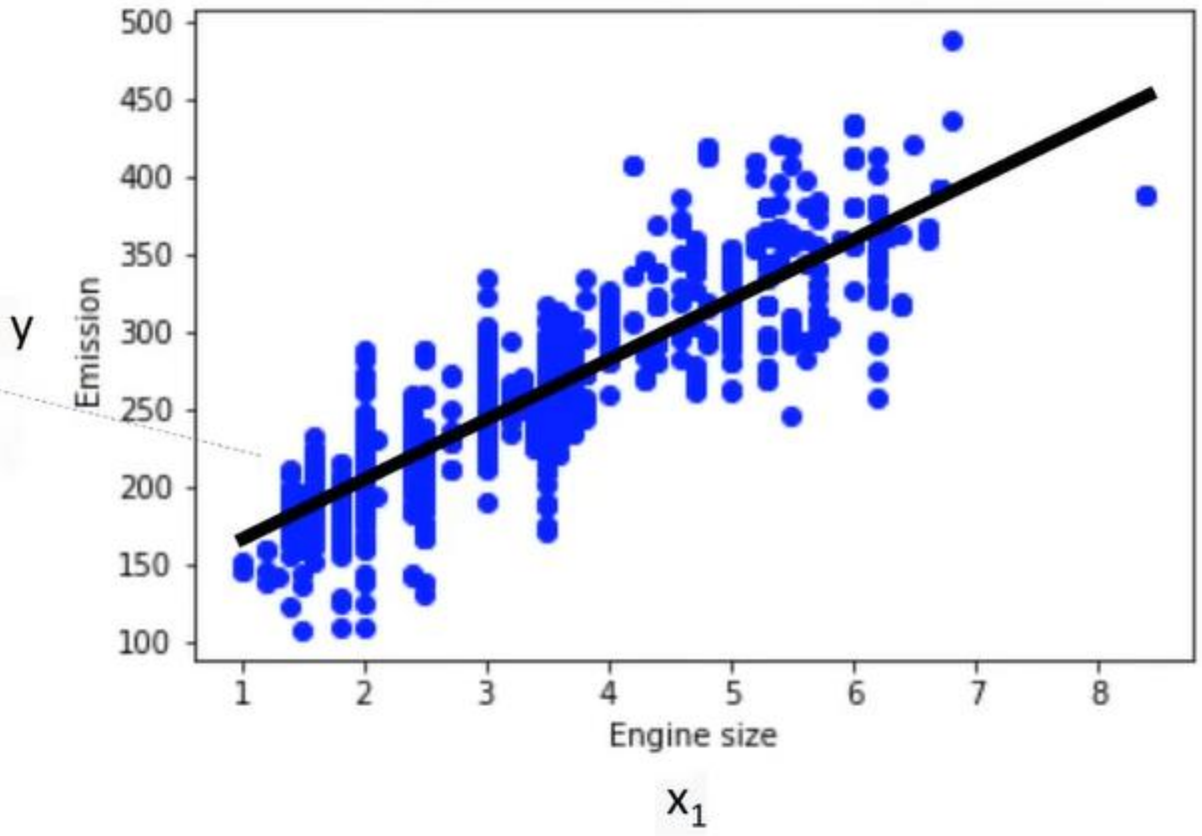
Linear Regression Model Representation

$$\hat{y} = w_0 + w_1 x$$

$$\hat{y} = \theta_0 + \theta_1 x_1$$

response variable

a single predictor



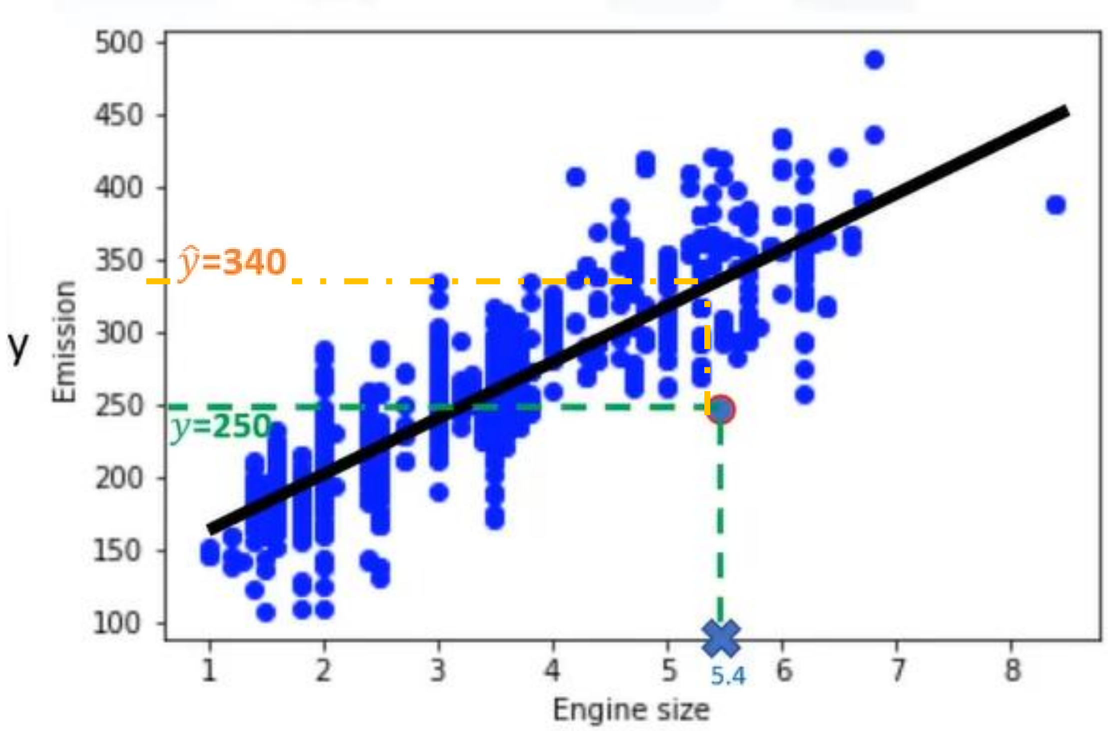
How to Find Best Fit Line?

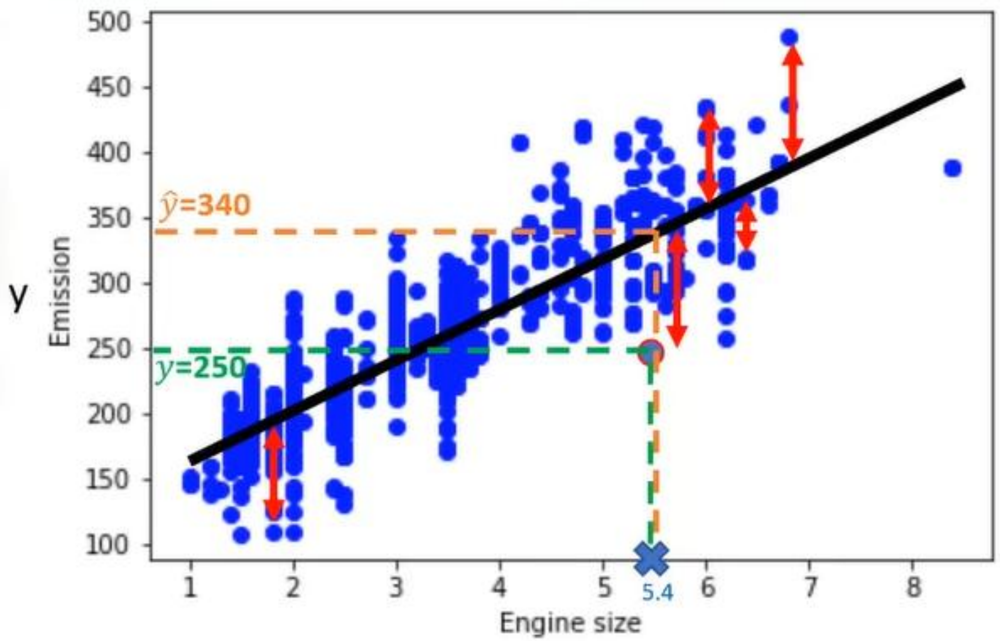
$x_1 = 5.4$ independent variable

$y = 250$ actual Co2 emission of x1

$\hat{y} = \theta_0 + \theta_1 x_1$

$\hat{y} = 340$ the predicted emission of x1





Error = y- $\hat{y}$

= 250 – 340

= -90

Objective: Minimize the Mean Squared Error

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Demo:
https://seeing-theory.brown.edu/regression-analysis/index.html#section1

Estimating the Parameter

$$\hat{y} = \theta_0 + \theta_1 x_1$$

$$\theta_1 = \frac{\sum_{i=1}^{s}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{s}(x_i - \bar{x})^2}$$

$$\bar{x} = (2.0 + 2.4 + 1.5 + \dots)/9 = 3.03$$

$$\bar{y} = (196 + 221 + 136 + \dots)/9 = 226.22$$

	ENGINESIZE	CYLINDERS	FUELCONSUMPTION_COMB	CO2EMISSIONS
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3	3.5	6	11.1	255
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5	3.5	6	10.0	230
6	3.5	6	10.1	232
7	3.7	6	11.1	255
8	3.7	6	11.6	267

X_1 y

$$\theta_1 = \frac{(2.0 - 3.03)(196 - 226.22) + (2.4 - 3.03)(221 - 226.22) + \dots}{(2.0 - 3.03)^2 + (2.4 - 3.03)^2 + \dots}$$

$$\theta_1 = 39$$

$$\theta_0 = \bar{y} - \theta_1 \bar{x}$$

$$\theta_0 = 226.22 - 39 * 3.03$$

$$\theta_0 = 125.74$$

$$\hat{y} = 125.74 + 39 x_1$$

Predicting with Linear Regression

	ENGINESIZE	CYLINDERS	FUELCONSUMPTION_COMB	CO2EMISSIONS
0	2.0	4	8.5	196
1	2.4	4	9.6	221
2	1.5	4	5.9	136
3	3.5	6	11.1	255
4	3.5	6	10.6	244
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6	3.5	6	10.1	232
7	3.7	6	11.1	255
8	3.7	6	11.6	267
9	2.4	4	9.2	?

$$\hat{y} = 125.74 + 39x_1$$

$$Co2Emission = \theta_0 + \theta_1\ EngineSize$$

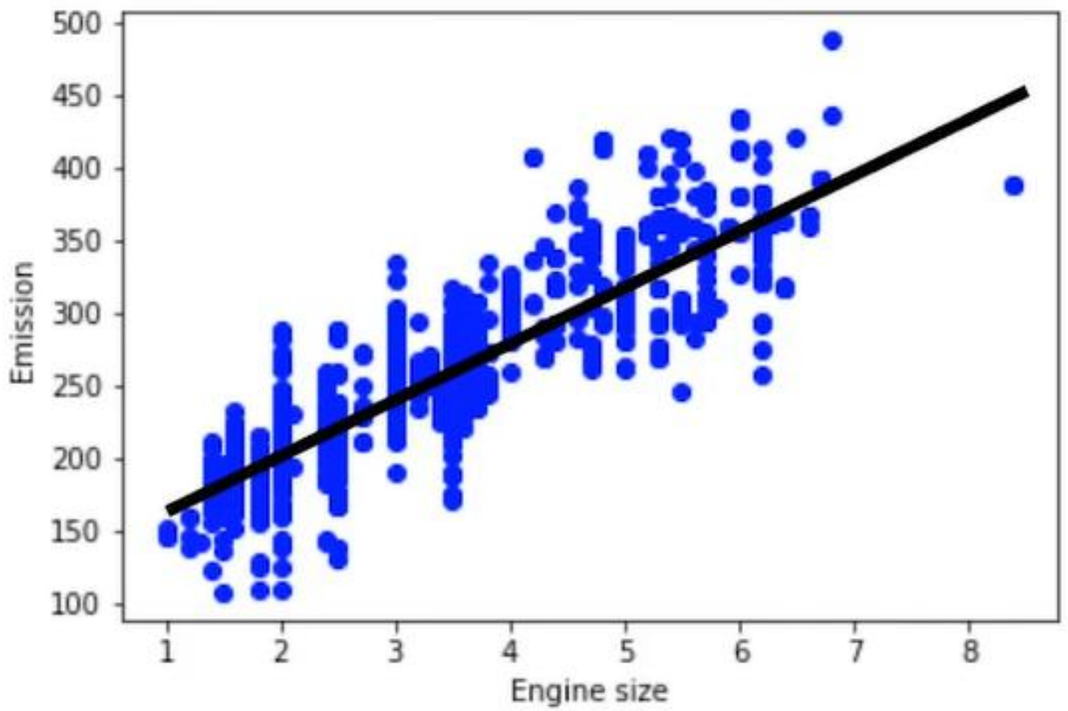
$$Co2Emission = 125 + 39\ EngineSize$$

$$Co2Emission = 125 + 39 \times 2.4$$

$$Co2Emission = 218.6$$

Pros of Linear Regression Model

- Very fast
- No parameter tuning
- Easy to understand, and highly interpretable



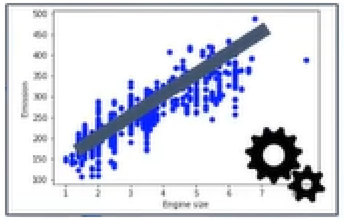
Model Evaluation in Regression Models

Train Test Split Evaluation Approach:

	ENGINESIZE	CYLINDERS	FUELCONSUMPTION_COMB	CO2EMISSIONS
0	2.0	4	8.5	196
1	2.4	4	9.6	221
2	1.5	4	5.9	136
3	3.5	6	11.1	255
4	3.5	6	10.6	244
5	3.5	6	10.0	230
6	3.5	6	10.1	232
7	3.7	6	11.1	255
8	3.7	6	11.6	267
9	2.4	4	9.2	212

Train

Test



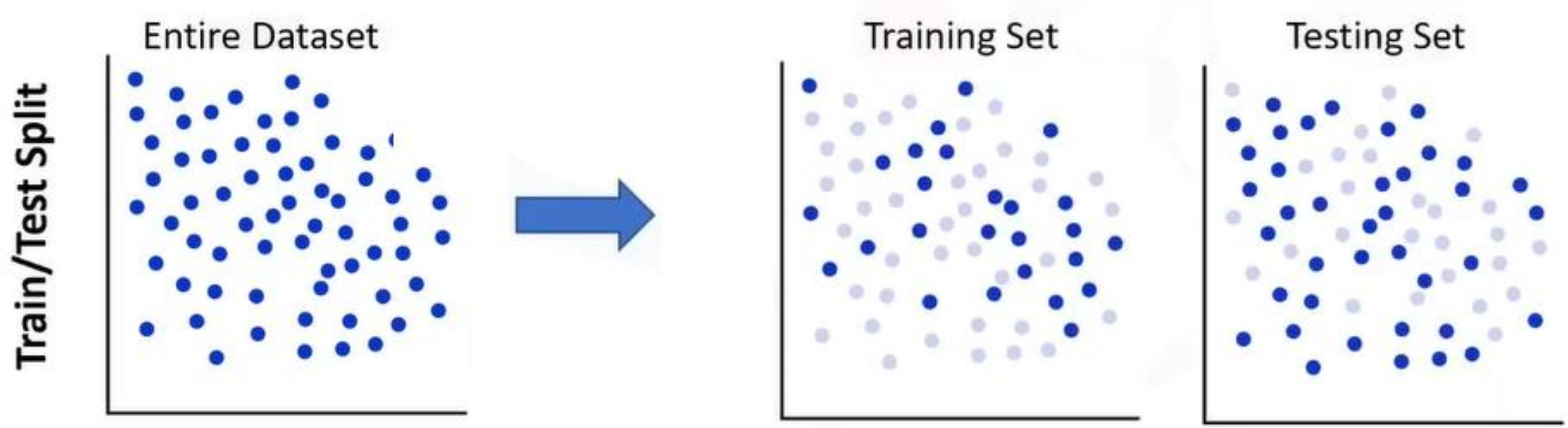
Actual Values

	Prediction
6	234
7	256
8	267
9	210

Compare

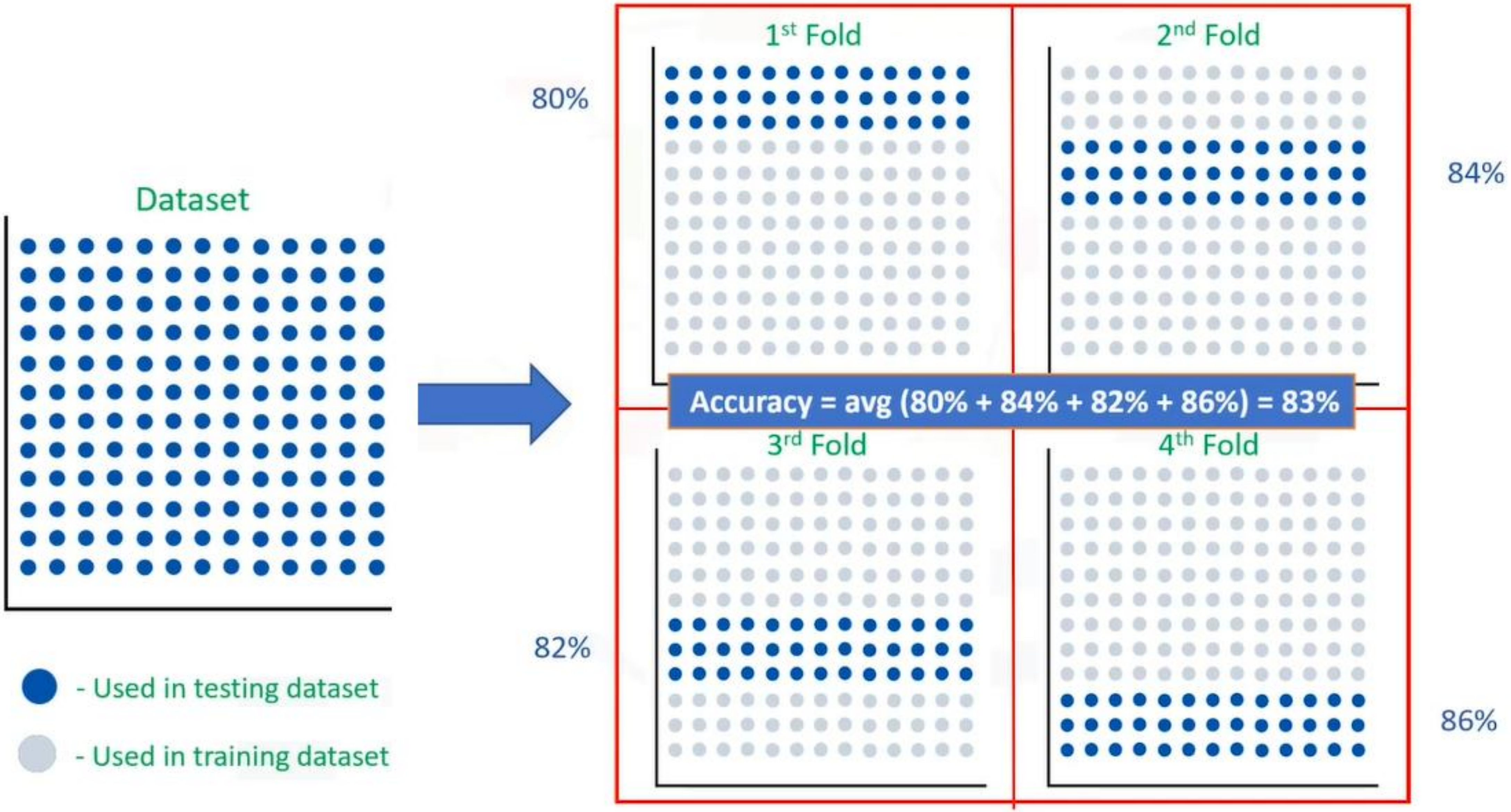
BREAK
15MINS

Train/Test Split Evaluation Approach



- Mutually Exclusive.
- More accurate evaluation on out-of-sample accuracy.
- Highly dependent on which datasets the data is trained and tested.

K Fold Cross-Validation Approach



Evaluation Metric in Regression Models

	ENGINESIZE	CYLINDERS	FUELCONSUMPTION_COMB	CO2EMISSIONS
0	2.0	4	8.5	196
1	2.4	4	9.6	221
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6	3.5	6	10.1	232
7	3.7	6	11.1	255
8	3.7	6	11.6	267
9	2.4	4	9.2	212

Test

y

Actual values

$$Error = \frac{(232 - 234) + (255 - 256) + \ \dots}{4}$$

$$Error = \frac{1}{n}\sum_{j=1}^{n}\left|y_j - \hat{y}_j\right|$$

$\hat{y}$

	Prediction
6	234
7	256
8	267
9	210

Predicted values

- MAE ★
- MSE ★
- RMSE ★
- ...

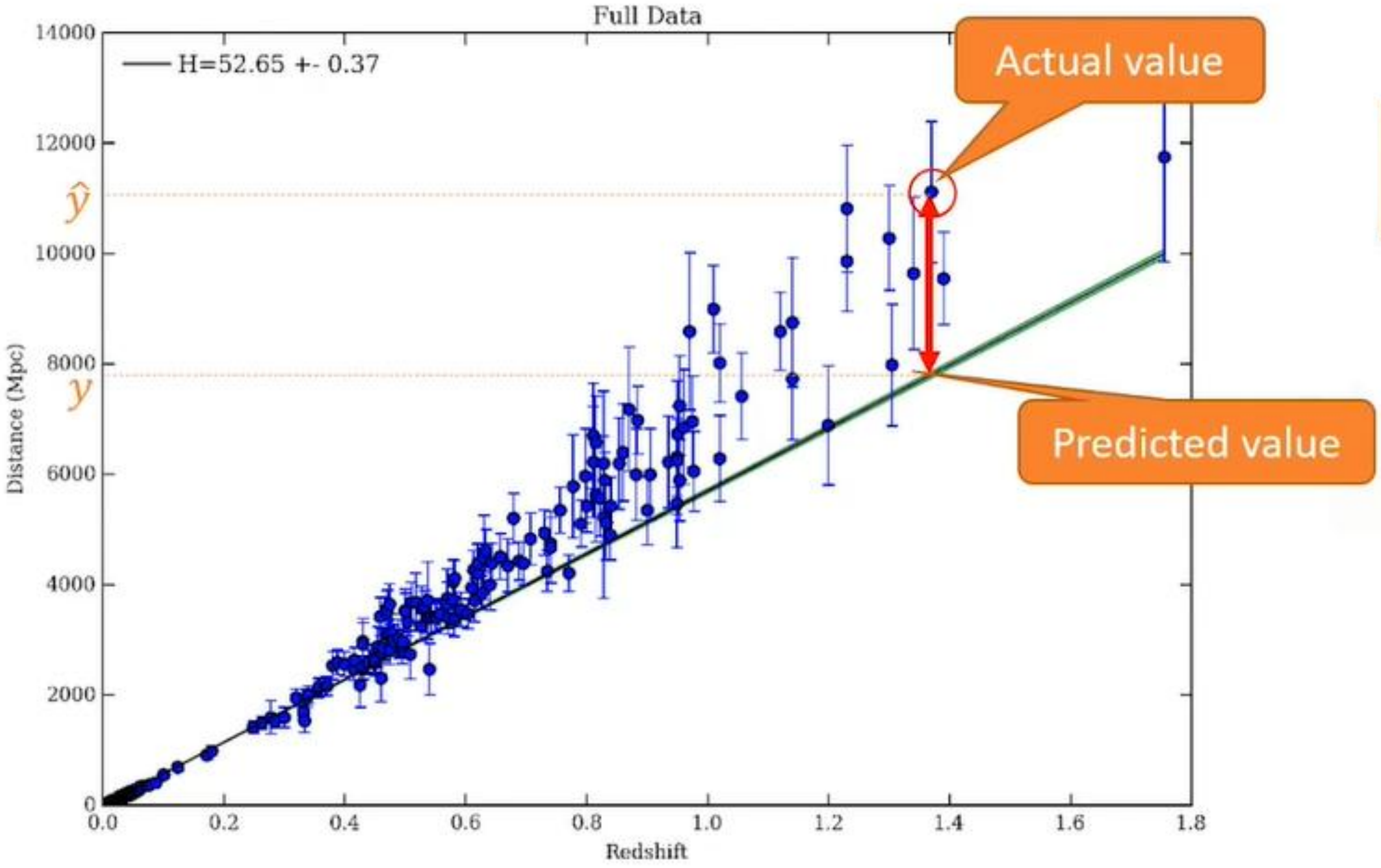
Scikit-learn.metrics

Regression metrics

See the Regression metrics section of the user guide for further details.

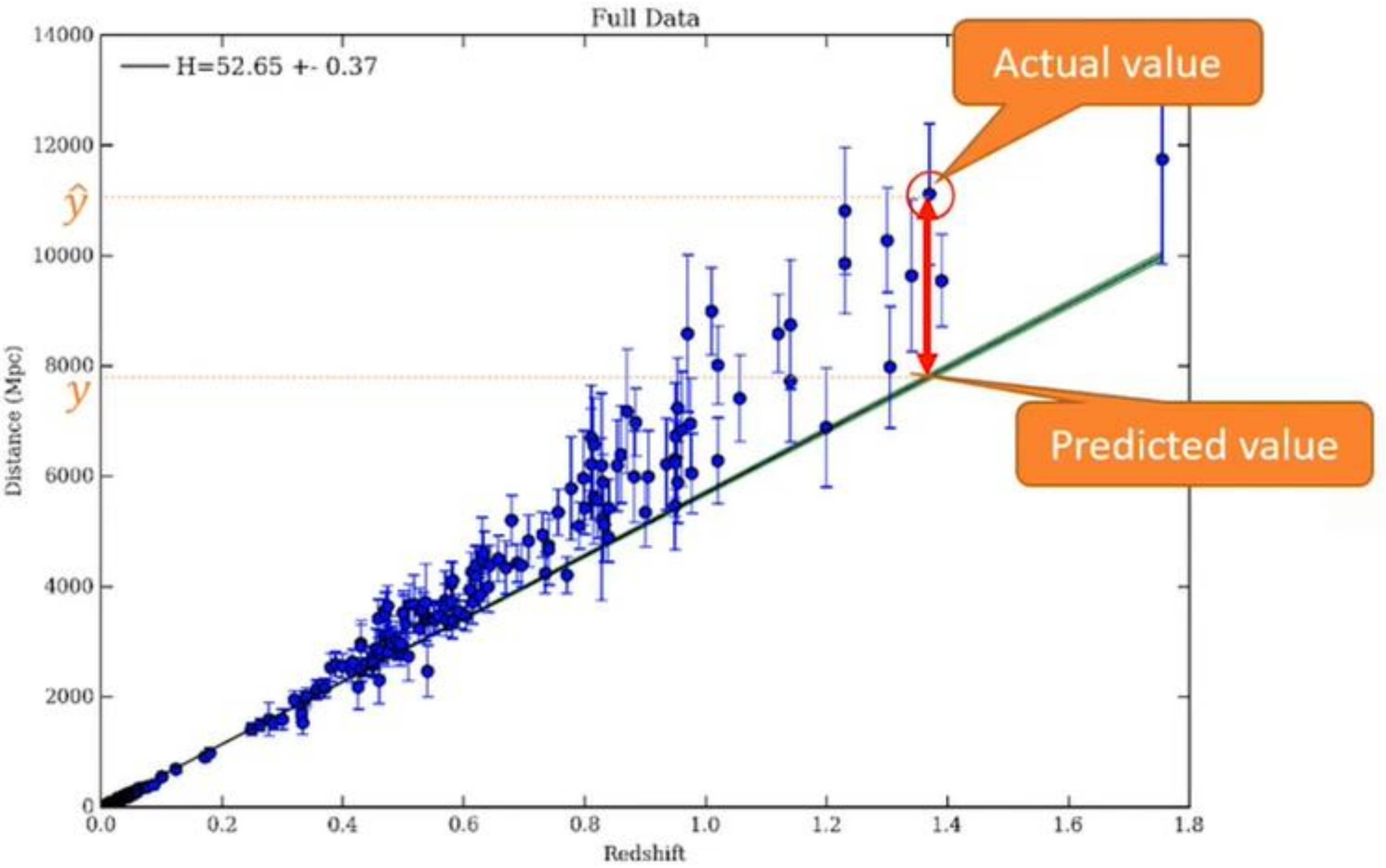
metrics.explained_variance_score (y_true, y_pred)	Explained variance regression score function
metrics.max_error (y_true, y_pred)	max_error metric calculates the maximum residual error.
metrics.mean_absolute_error (y_true, y_pred)	Mean absolute error regression loss
metrics.mean_squared_error (y_true, y_pred[, …])	Mean squared error regression loss
metrics.mean_squared_log_error (y_true, y_pred)	Mean squared logarithmic error regression loss
metrics.median_absolute_error (y_true, y_pred)	Median absolute error regression loss
metrics.r2_score (y_true, y_pred[, …])	R^2 (coefficient of determination) regression score function.

Evaluation Metric in Regression Models



Error: measure of how far the data is from the fitted regression line.

Evaluation Metric in Regression Models



$$MAE = \frac{1}{n}\sum_{j=1}^{n}|y_j - \hat{y}_j|$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$\mathrm{R}MSE = \sqrt{\frac{1}{n}\sum_{j=1}^{n}\left(y_j - \hat{y}_j\right)^2}$$

$$RAE = \frac{\sum_{j=1}^{n}|y_j - \hat{y}_j|}{\sum_{j=1}^{n}|y_j - \bar{y}|}$$

$$RSE = \frac{\sum_{j=1}^{n}\left(y_j - \hat{y}_j\right)^2}{\sum_{j=1}^{n}\left(y_j - \bar{y}\right)^2}$$

$$R^2 = 1 - RSE$$

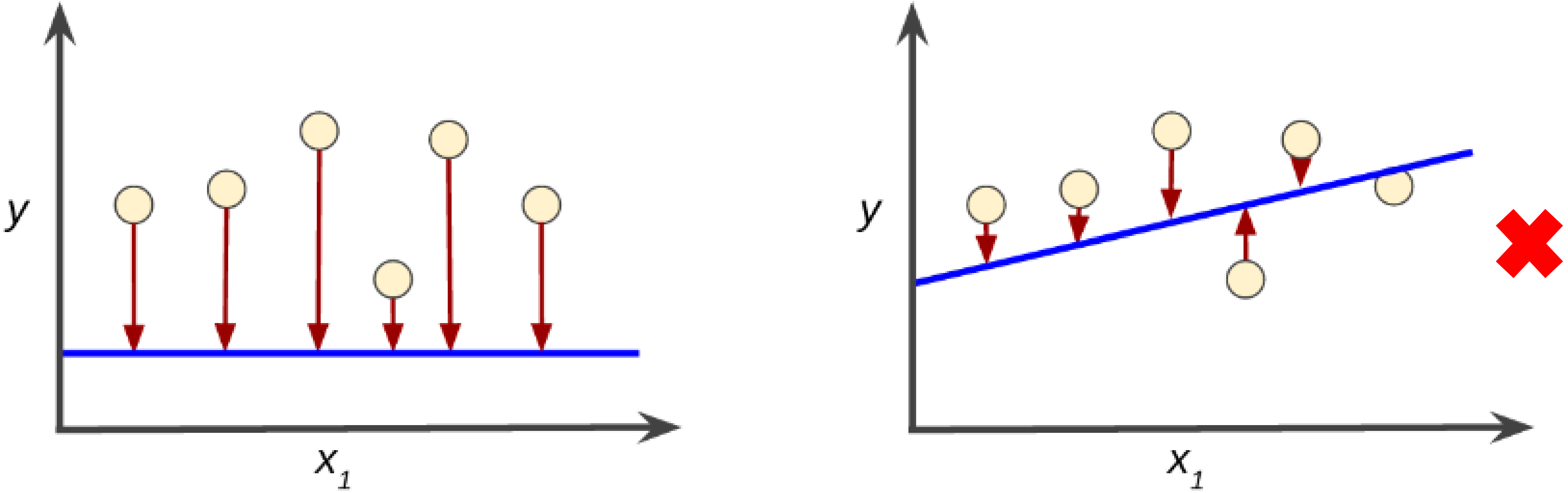
Mean Square Error

Objective: Minimize the Mean Squared Error

This is Called as Empirical Risk Minimization

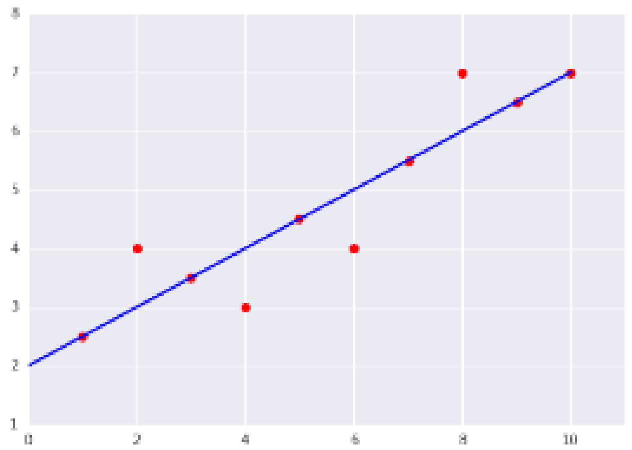
$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

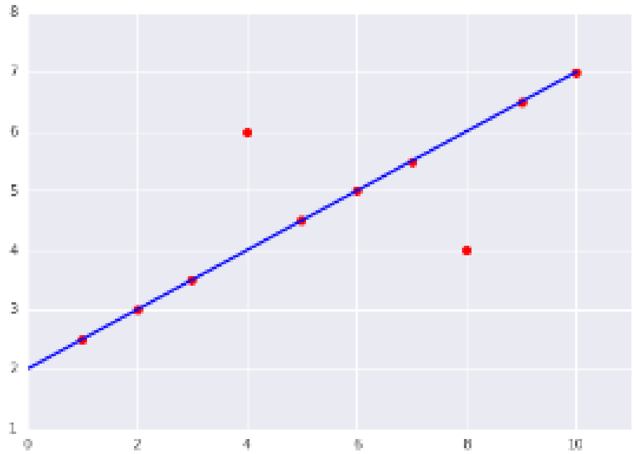
In Which Plot is the Loss Minimum?



Trivia Time

Consider the following two plots:





Explore the options below.

Which of the two data sets shown in the preceding plots has the higher Mean Squared Error (MSE)?

The dataset on the right.

$$MSE = \frac{0^2 + 0^2 + 0^2 + 2^2 + 0^2 + 0^2 + 0^2 + 2^2 + 0^2 + 0^2}{10} = 0.8$$

The dataset on the left.

$$MSE = \frac{0^2 + 1^2 + 0^2 + 1^2 + 0^2 + 1^2 + 0^2 + 1^2 + 0^2 + 0^2}{10} = 0.4$$

Multiple Regression Models

X: Independent variable

Y: Dependent variable

	ENGINESIZE	CYLINDERS	FUELCONSUMPTION_COMB	CO2EMISSIONS
0	2.0	4	8.5	196
1	2.4	4	9.6	221
2	1.5	4	5.9	136
3	3.5	6	11.1	255
4	3.5	6	10.6	244
5	3.5	6	10.0	230
6	3.5	6	10.1	232
7	3.7	6	11.1	255
8	3.7	6	11.6	267
9	2.4	4	9.2	?

$$Co2\ Em = \theta_0 + \theta_1 Engine\ size + \theta_2 Cylinders + \ \ldots$$

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ \ldots + \theta_n x_n$$

$$\hat{y} = \theta^T X$$

$$\theta^T = [\theta_0, \theta_1, \theta_2, \ldots] \qquad X = \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ \ldots \end{bmatrix}$$

Estimating Parameters of Multiple Regression Models

	ENGINESIZE	CYLINDERS	FUELCONSUMPTION_COMB	CO2EMISSIONS
0	2.0	4	8.5	196
1	2.4	4	9.6	221
2	1.5	4	5.9	136
3	3.5	6	11.1	255
4	3.5	6	10.6	244
5	3.5	6	10.0	230
6	3.5	6	10.1	232
7	3.7	6	11.1	255
8	3.7	6	11.6	267

$\hat{y} = \theta^T X$

$\hat{y}_i = 140$ the predicted emission of x_i

$y_i = 196$ actual value of x_i

$y_i - \hat{y}_i = 196 - 140 = 56$ residual error

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

- How to estimate θ?
  - Ordinary Least Squares
    - Linear algebra operations
    - Takes a long time for large datasets (10K+ rows)
  - An optimization algorithm
    - Gradient Descent
    - Proper approach if you have a very large dataset

Making Predictions with Multiple Regression Models

	ENGINESIZE	CYLINDERS	FUELCONSUMPTION_COMB	CO2EMISSIONS
0	2.0	4	8.5	196
1	2.4	4	9.6	221
2	1.5	4	5.9	136
3	3.5	6	11.1	255
4	3.5	6	10.6	244
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6	3.5	6	10.1	232
7	3.7	6	11.1	255
8	3.7	6	11.6	267
9	2.4	4	9.2	?

$$\hat{y} = \theta^T X$$

$$\theta^T = [125, 6.2, 14, \ldots]$$

$$\hat{y} = 125 + 6.2x_1 + 14x_2 +$$

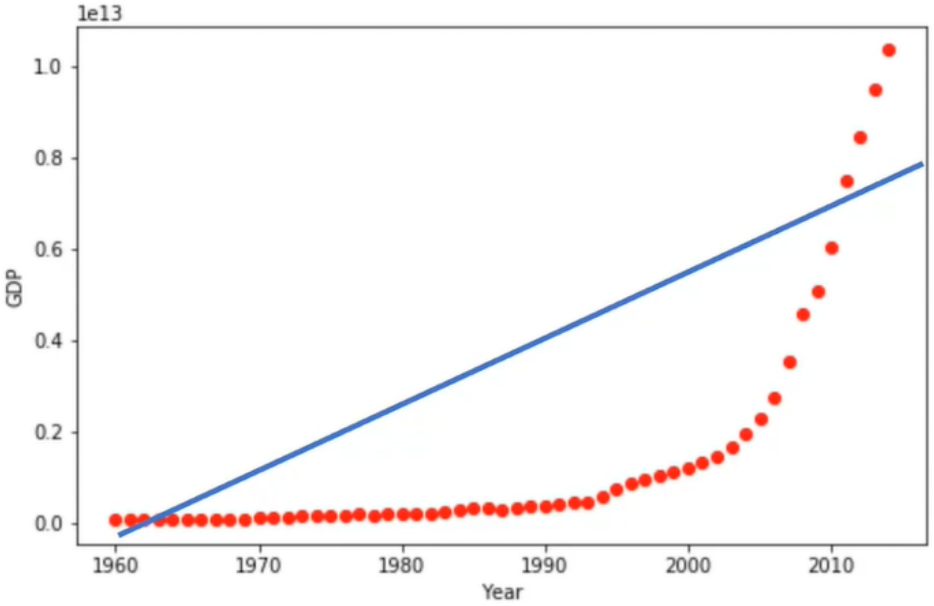
$$Co2Em = 125 + 6.2EngSize + 14\text{ Cylinders} + \ldots$$

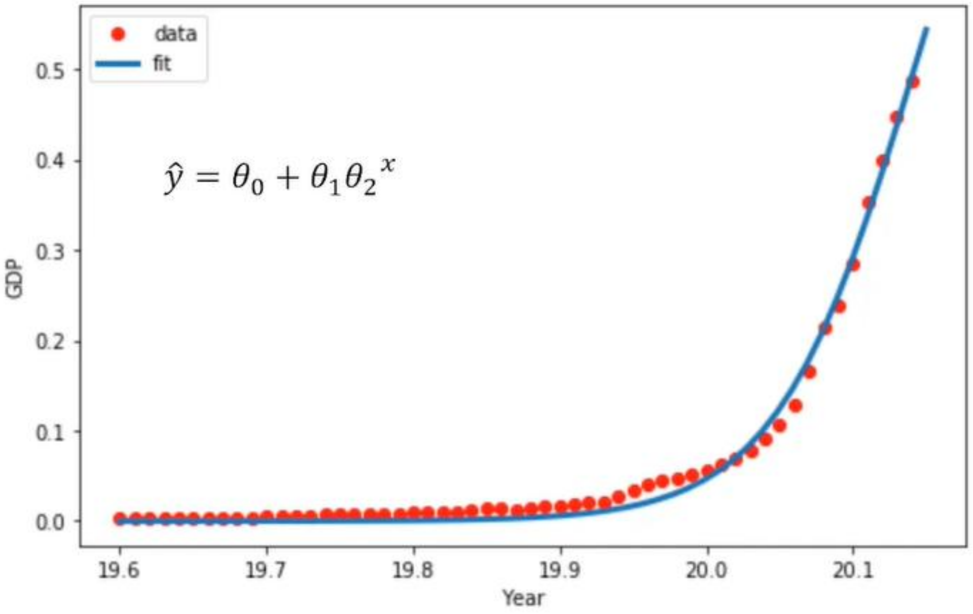
$$Co2Em = 125 + 6.2 \times 2.4 + 14 \times 4 + \ldots$$

$$Co2Em = 214.1$$

Non-Linear Regression Models

	Year	Value
0	1960	5.918412e+10
1	1961	4.955705e+10
2	1962	4.668518e+10
3	1963	5.009730e+10
4	1964	5.906225e+10
5	1965	6.970915e+10
6	1966	7.587943e+10
7	1967	7.205703e+10
8	1968	6.999350e+10
9	1969	7.871882e+10
...	.....	....





Non-Linear Regression

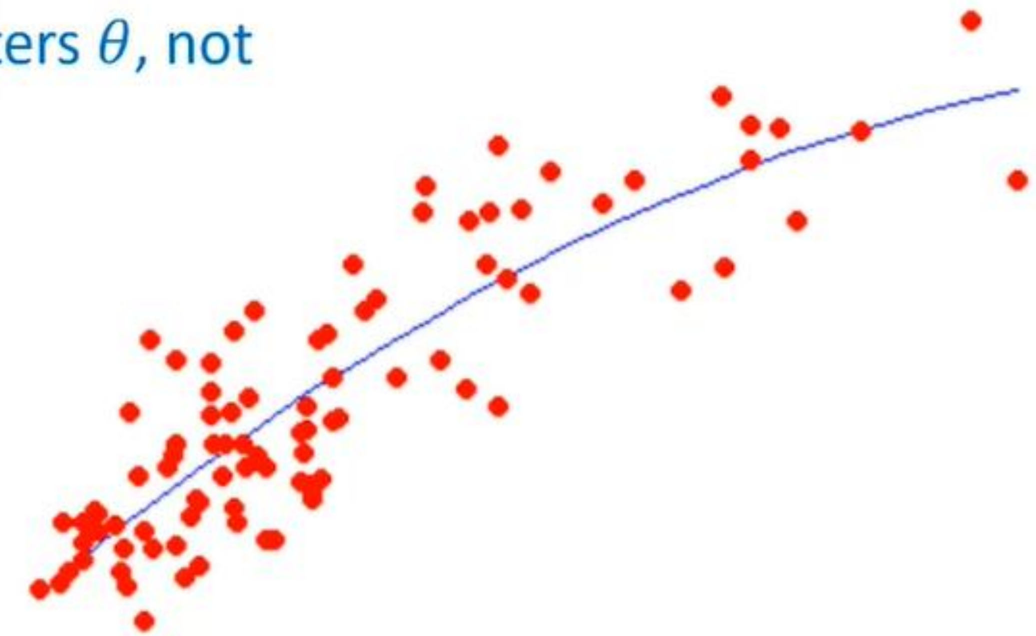
- To model non-linear relationship between the dependent variable and a set of independent variables
- $\hat{y}$ must be a non-linear function of the parameters θ, not necessarily the features x

$$\hat{y} = \theta_0 + {\theta_2}^2 x$$

$$\hat{y} = \theta_0 + \theta_1 {\theta_2}^x$$

$$\hat{y} = \log(\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3)$$

$$\hat{y} = \frac{\theta_0}{1 + {\theta_1}^{(x-\theta_2)}}$$

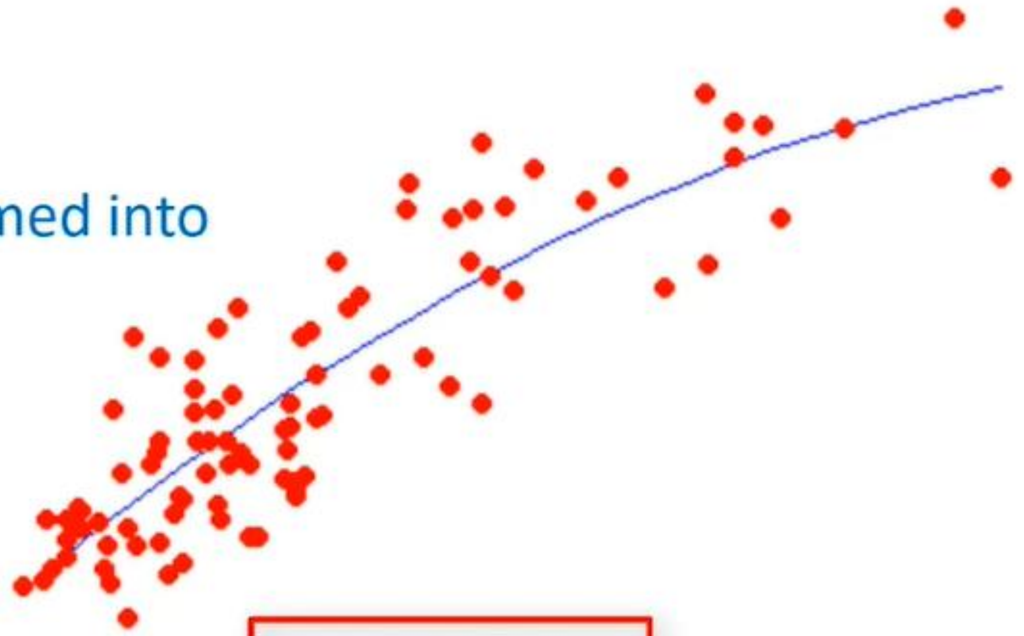


Polynomial Regression

- Some curvy data can be modeled by a polynomial regression
- For example:

$$\hat{y} = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$$

- A polynomial regression model can be transformed into linear regression model.

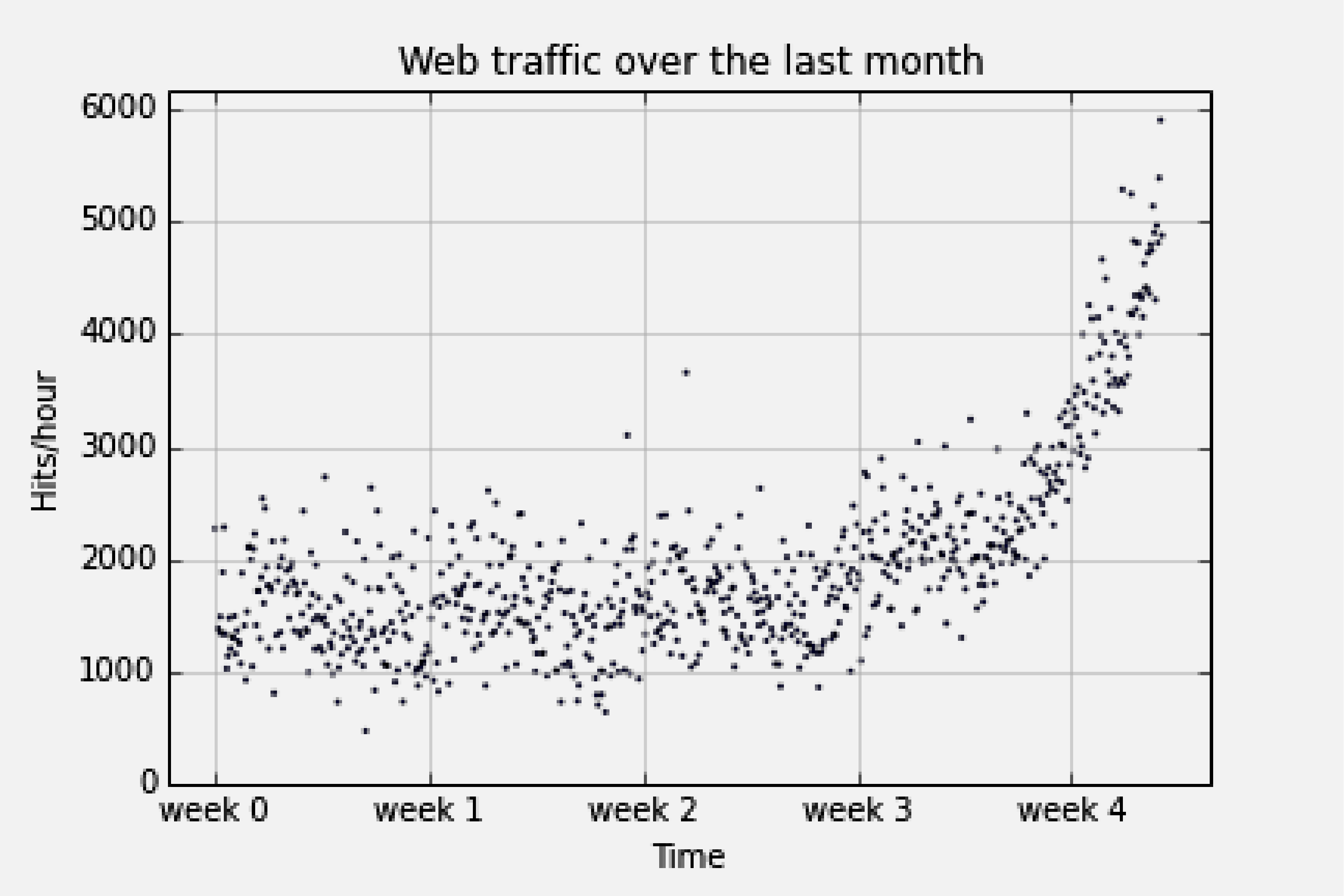


$x_1 = x$
$x_2 = x^2$
$x_3 = x^3$

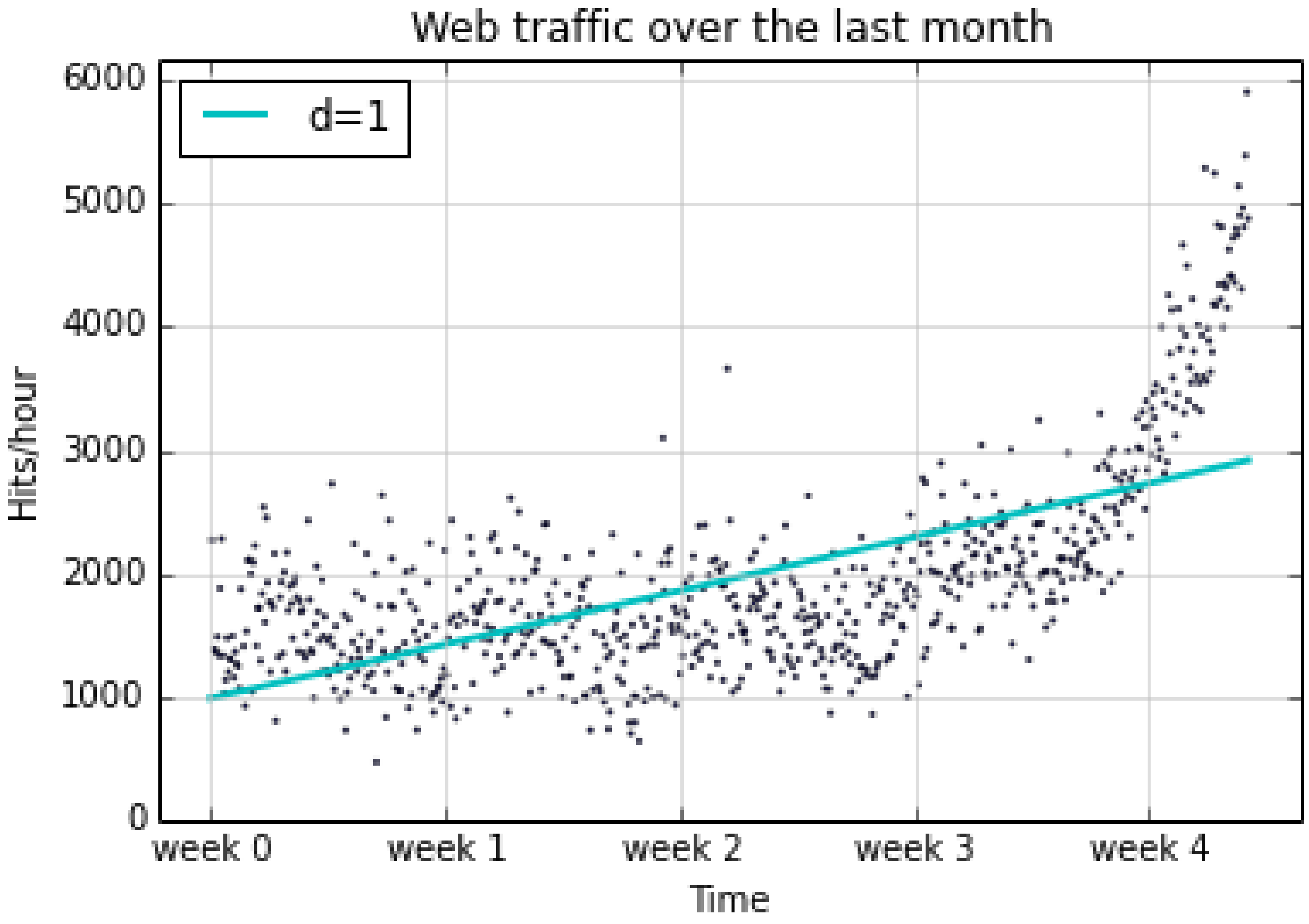
$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$ → Multiple linear regression → Least Squares

Minimizing the sum of the squares of the differences between y and $\hat{y}$

Example: Web Traffic Data

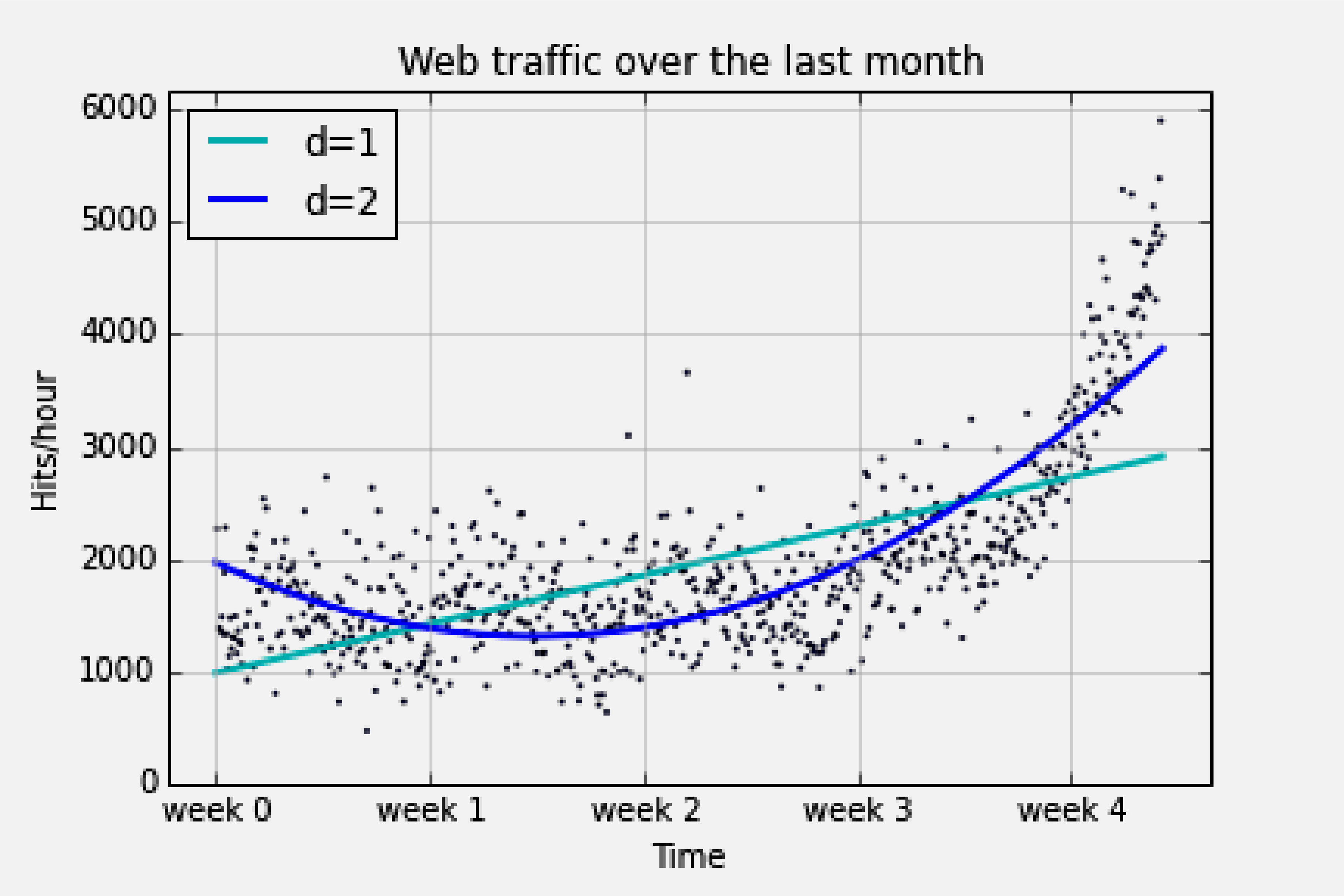


1D Poly Fit

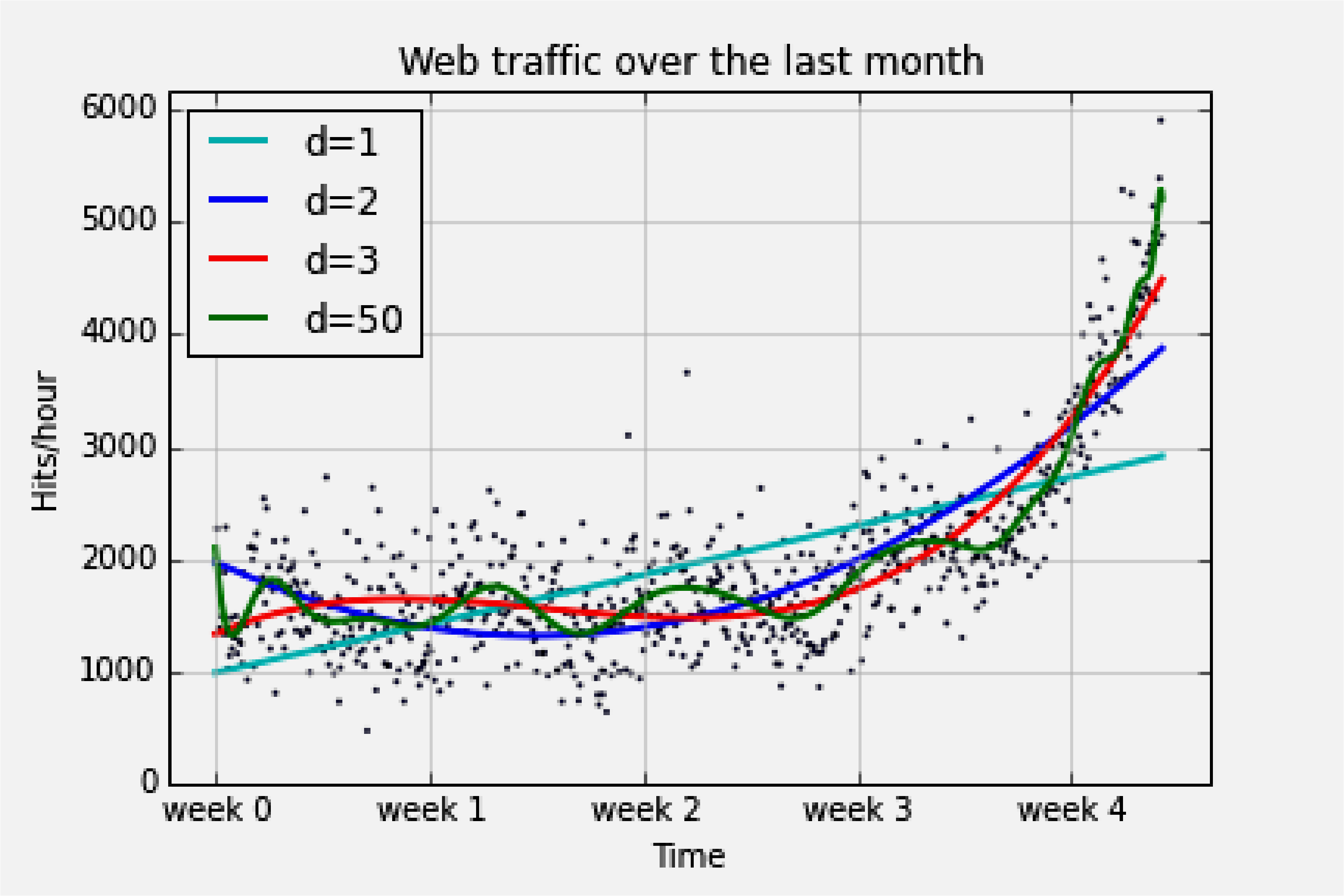


Example of too much “bias” → underfitting

Example: 1D and 2D Poly Fit



Example: 1D Ploy Fit

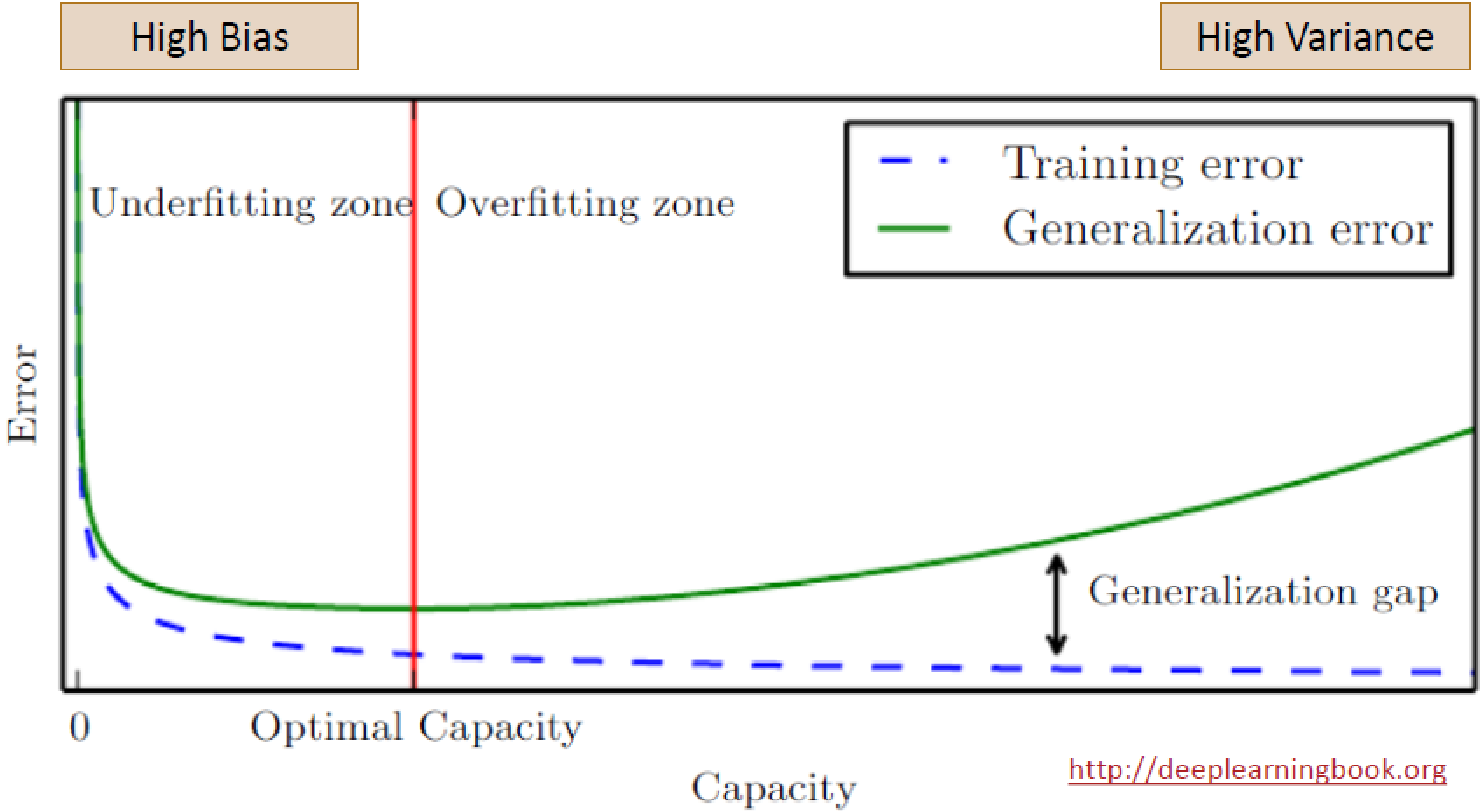


Example of too much “variance” → overfitting

BREAK
15MINS

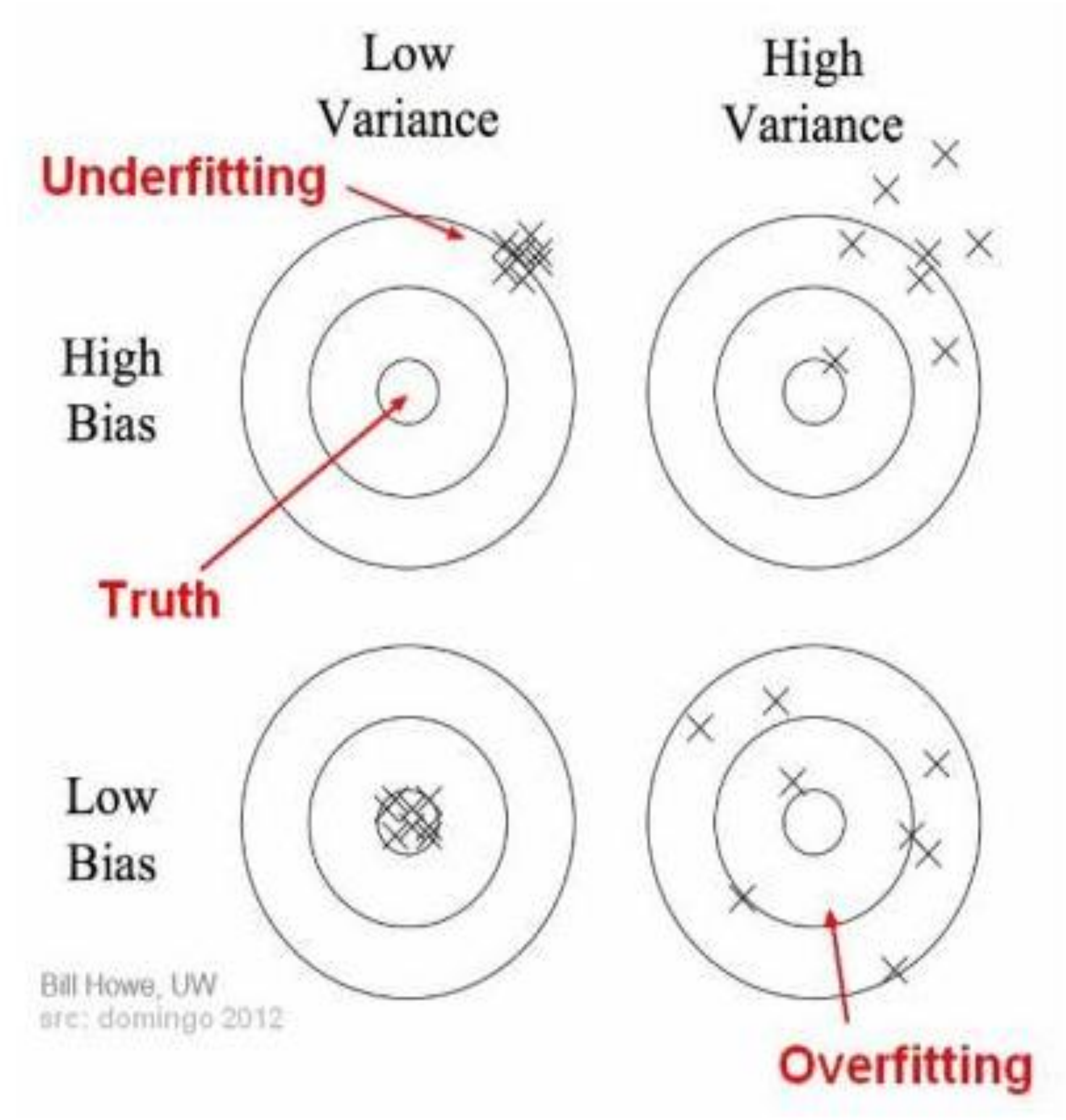
Generalization (or Test) Error

- The central challenge in ML is to perform well on new, previously unseen inputs

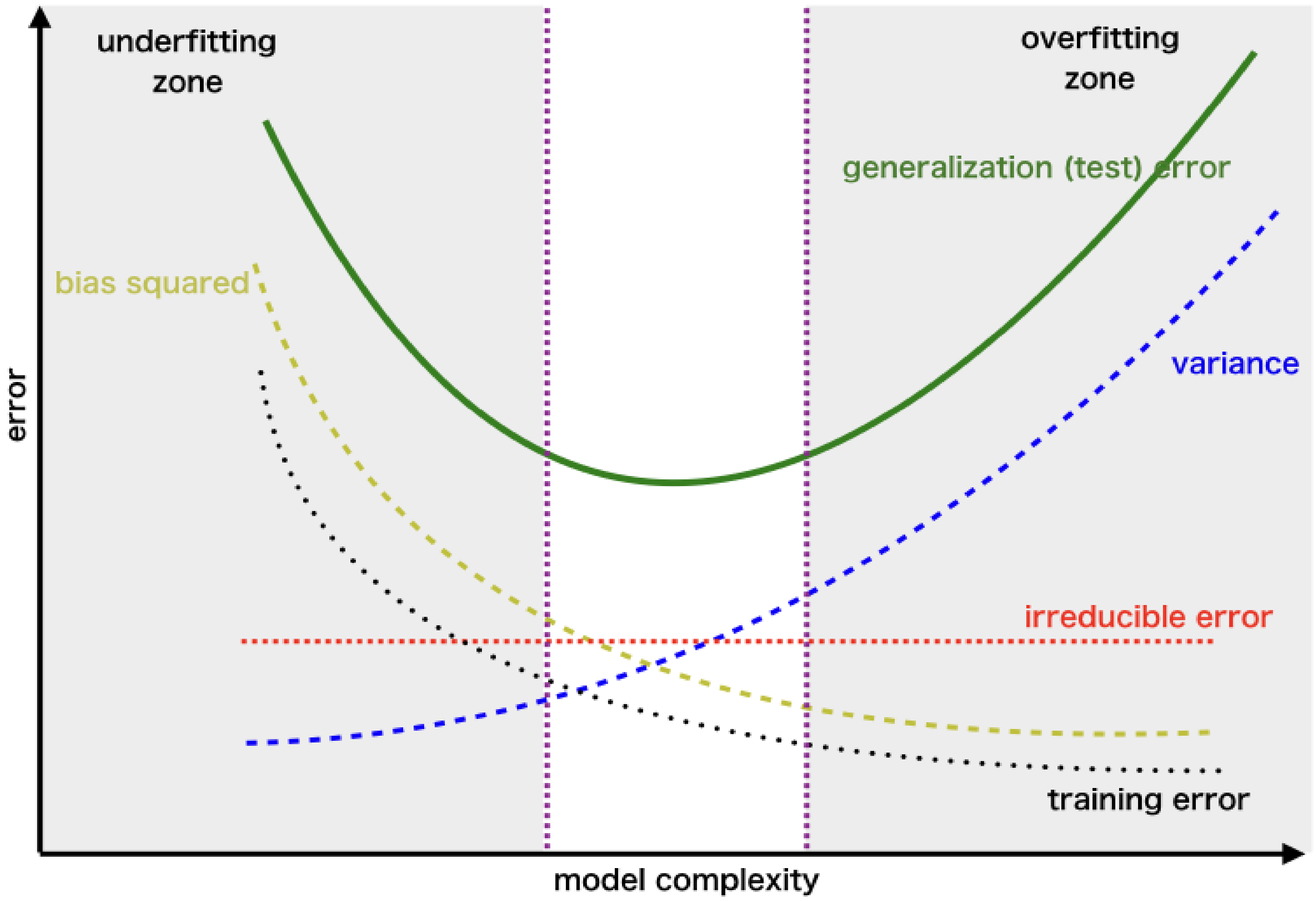


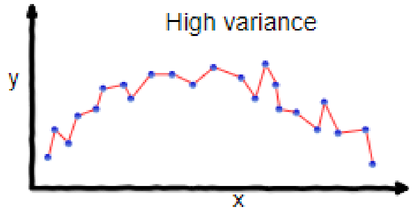
http://deeplearningbook.org

Bias-Variance Tradeoff

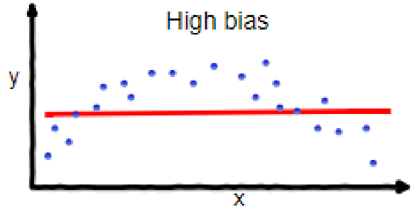


Bias-Variance Tradeoff

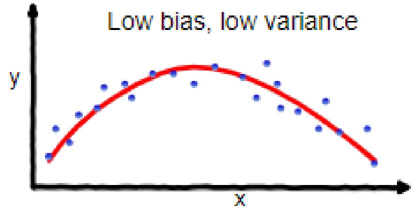




overfitting



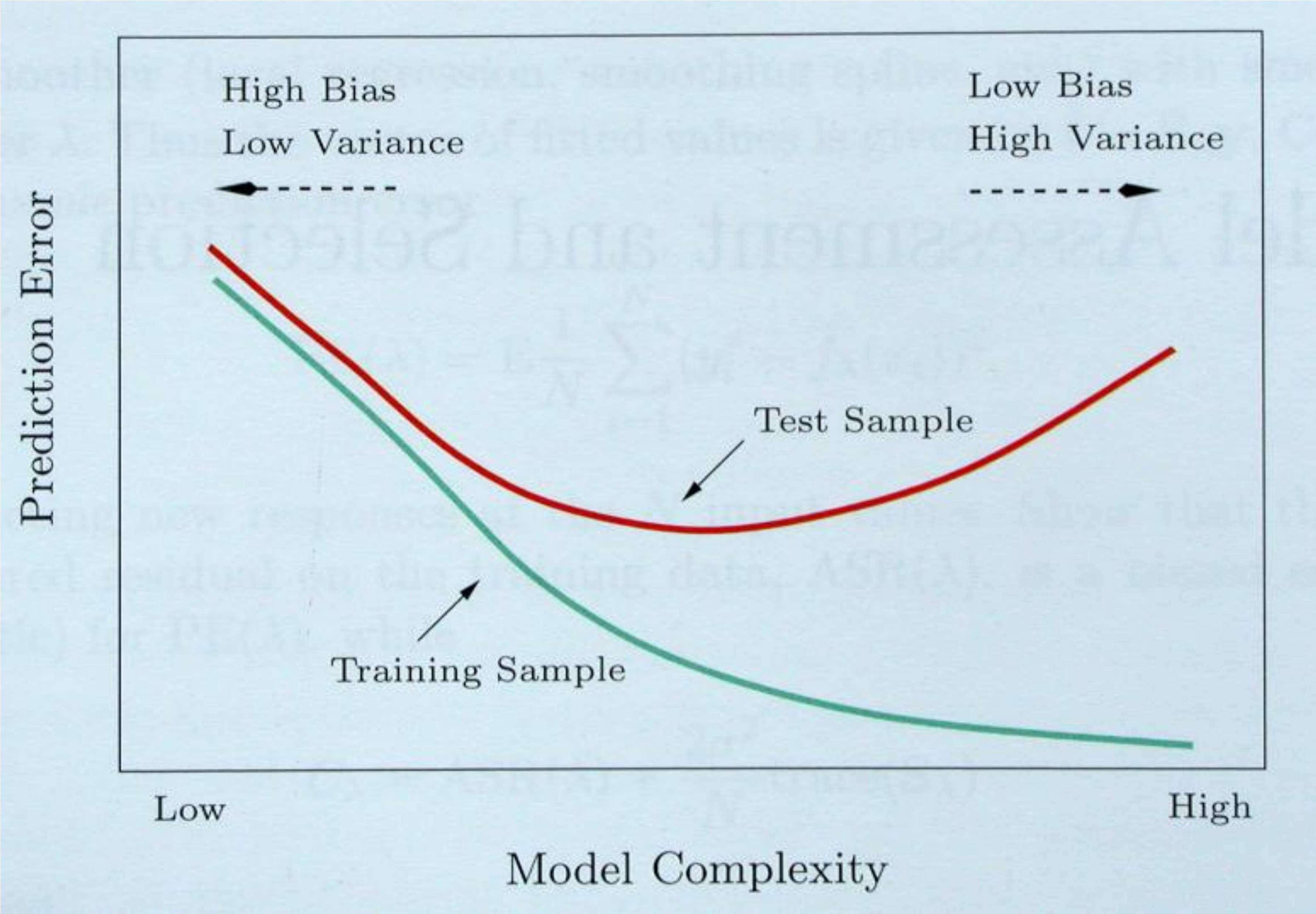
underfitting



Good balance

Image Source

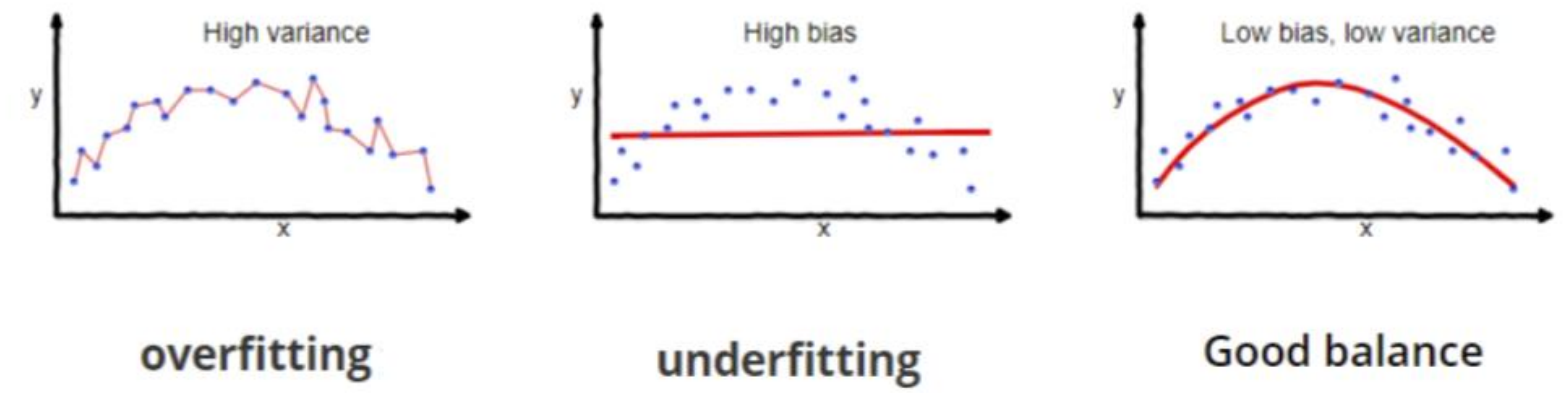
Bias-Variance Tradeoff



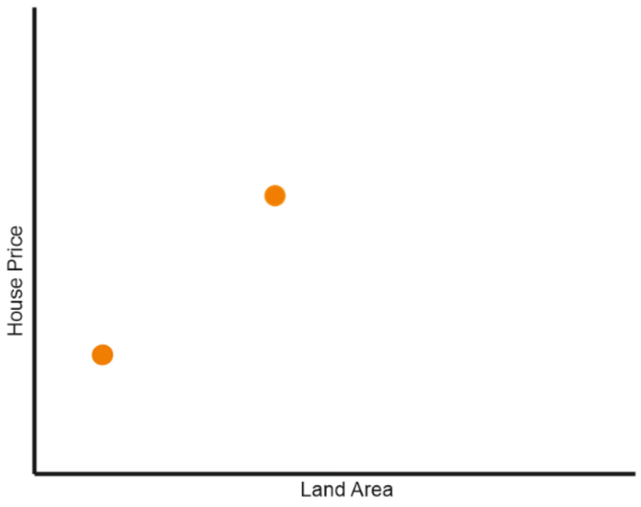
Regularization

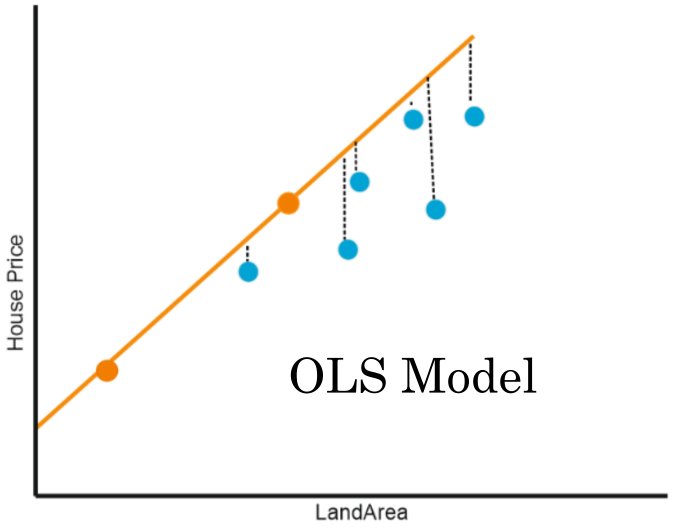
- Complex models (lots of parameters) are often prone to overfitting

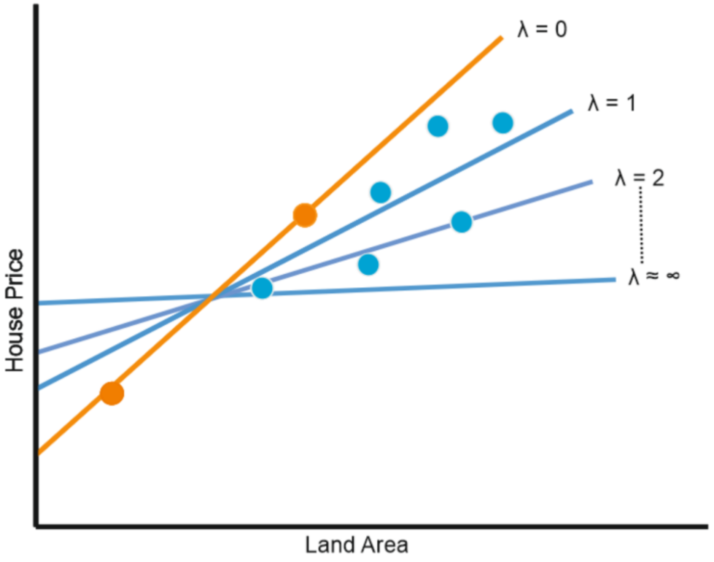
- Overfitting can be reduced by imposing a constraint on the overall magnitude of the parameters (i.e., by including coefficients as part of the optimization process)

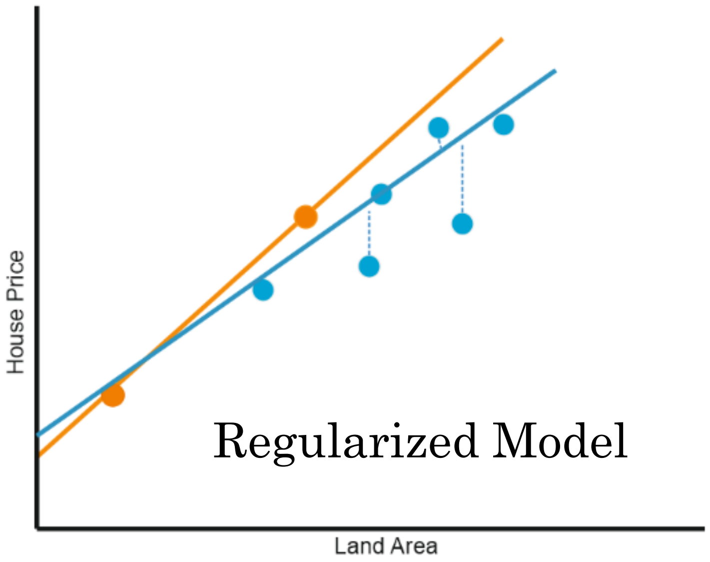


Regularization









Regularization

- Two common types of regularization in linear regression:

  ❖ L_2 regularization (a.k.a. ridge regression). Find **w** which minimizes: $\sum_{j=1}^{N}(y_j - \sum_{i=0}^{d} w_i \cdot x_i)^2 + \lambda \sum_{i=1}^{d} w_i^2$

    - λ is the regularization parameter: bigger λ imposes more constraint

  ❖ L_1 regularization (a.k.a. lasso). Find **w** which minimizes: $\sum_{j=1}^{N}(y_j - \sum_{i=0}^{d} w_i \cdot x_i)^2 + \lambda \sum_{i=1}^{d} |w_i|$

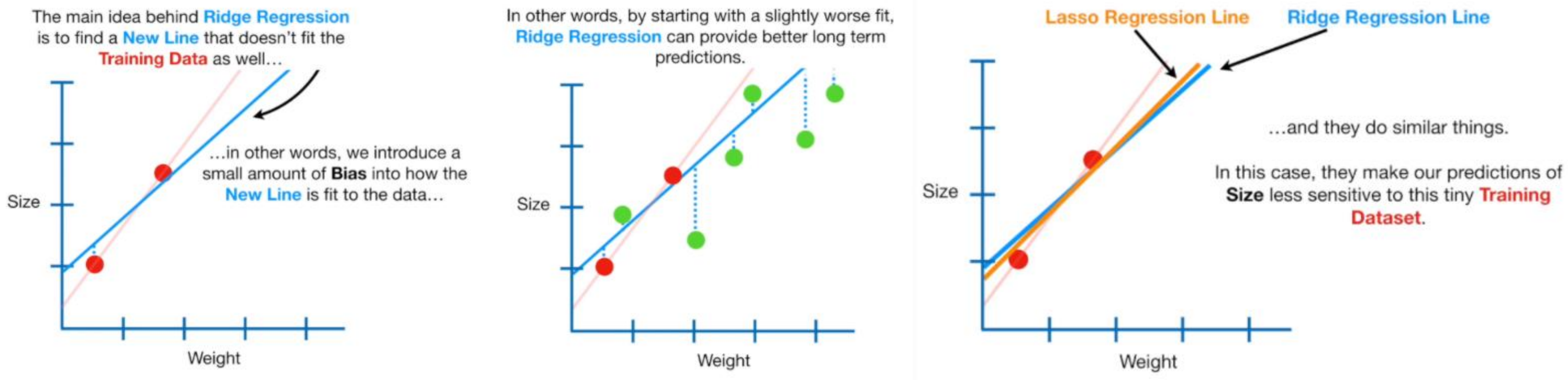


Image Source: https://www.andreaperlato.com/theorypost/ridge-and-lasso-regression/

Other Regression-Based Models

Home Work:
https://scikit-learn.org/stable/modules/linear_model.html

Mentimeter Quiz

Mentimeter

Get Ready to Test Your Knowledge

Thank You for Attention